Note on Latin Grammar

Jinyuan Wu

August 24, 2026

Chapter 1

Introduction

1.1 The language and the speakers

Latin was the language of the Romans and the official language of both the Roman Republic and the Roman Empire, and hence the official language of the Catholic Church, which was *the* church for the Western Roman Empire. The international nature of the Roman Empire made Latin the international language around the Mediterranean Sea at that time – indeed, *Mare Nostrum* 'our sea' in Latin, and its importance in science, arts, law, religion, and literature lent it more than one thousand of years of life as a common literary language and a sacred language in western Europe after the collapse of the Western Roman Empire and the emergence of the Romance language family.

As recently as the nineteen century, Latin was still fluently used by scholars and in the Catholic Mass. A decline in the popularity of Latin was observed after that, consistent with the rapid development of English (initially also French and German and sometimes Russian) as the language of rapidly developing natural sciences, largely replacing the status of Latin as a scholar language. After Vatican allowed vernacular languages being used in liturgies, Latin also largely lose its position in the daily use in the Catholic church.

Latin today is still revered as *a* classical language that has deeply shaped at least the Western world, but it has long lost the prestige of *the* classical language that every decent individual is to learn once in their life. Still, as an important silhouette of what early Indo-European languages looked like, as one of the earliest languages receiving systematic grammatical analysis, and as a language that considerable deviates from its modern descendants in its grammar, Latin deserves its status as a model organism in modern linguistics.

1.2 Classification and history

1.2.1 Latin in Indo-European family

Latin belongs to the Indo-European (IE) family. Languages most closely genetically related to Latin include Faliscan, a language attested from seventh-sixth century BC, Lanuvian, and Praenestinian. These languages form the Latino-Faliscan family.

The Sabellic family, which consists of Umbrian, Oscan, and South Picene, was also spoken in ancient Italy. The exact relation between Sabellian and Latino-Faliscan is not completely settled: many postulate an Italic language family with Latino-Faliscan and Sabellian as its two branches.

Above the level of the Italic family, it has been proposed that Latin is genetically related to the Celtic branch.

1.2.2 Old Latin

Pre-Classical Latin had gone dramatic evolution. The earliest texts are still not fully deciphered today and the Romans themselves find it hard to read old agreements they signed with the Carthaginians (Leonhardt, 2013, p. 52). Indeed, late Old Latin texts had already shown changes that are also present in Vulgar Latin, which are however absent in Classical texts, suggesting archaisms in the standardization process which makes the Classical texts *appearing older* than they actually are, while late Old Latin texts are actually *later* in certain linguistic features when compared with idiomatic Classical Latin texts in the sense that they reflect innovations in the living language (§ 1.2.3).

Box 1.1: List of Old Latin traits

- *-os* as ending
- texts?

1.2.3 Emergence of Classical Latin

The road to standardization The formation of what is known today as "normal" Latin, i.e. the kind of Latin usually taught in Latin classes, follows the general pattern of linguistic standardization. The motivation of the process was closely connected to the formation of new identities, which, in the case of standardization of Latin, was the awareness that the Romans should focus on their own cultures and got some sort of spiritual independence from the Greeks: thus Virgil sought to replace *Odyssey* by his *Aeneid*, and not just to translate the former (Leonhardt, 2013, p. 66). The focus on the new identify triggered the so-called "first congress" phenomenon, in which broad discussion on what variety of a language is to be used and eventually the creation of a new literary language. In Rome, this was done by leading members of the upper society: Cicero mentioned linguistic problems in his discussions on rhetorics, Caesar was worried about irregularities of Latin inflectional morphology and proposed some spelling reforms to make homophonous endings more identifiable, and several other figures composed grammars of Latin (Leonhardt, 2013, pp. 60-61).

Certain traits of Classical Latin are also recognizably deliberately refined. The situation is the most telling when a feature is present in both pre-Classical texts (especially dramas, where the language is conceivably colloquial) and post-Classical texts but absent in Classical Latin, meaning that this feature was artificially suppressed in standardization (but likely had always stayed in oral uses). Instances include missing word final *-m* and *de* + noun phrase in place of the genitive case (Alkire and Rosen, 2010, § 2.4.4, § 8.3.1).

1.2.3.1 Classical Latin as a spoken language

Although Classical Latin underwent deliberate refinements, it was still a true living language in its time, namely around the first century. Evidence for this claim comes from both letters and oration, known to reflect the oral language according to literal evidence.

Cicero, for instance, was equally Classical in his popular oratory and in his written works, and the fact that public oration was to be understood by as many as possible clearly shows that Cicero's language was a real language that was actually spoken by at least Roman elites (Leonhardt, 2013, p. 71), which is even true for the language of legal procedures, commonly expected to be rather technical: although there were legal experts in ancient Rome, the heads of tribunals were often not from their ranks, and to capture their attentions the everyday language had to be used (Clackson, 2011, pp. 385-386,464-465). As for letters, he explicitly mentioned in his letter to her close friend that "as for letters, we weave them out of the language of everyday" (Clackson, 2011, p. 428).

Of course, both letters and public speeches, in all cultures, undergo deliberate refinements, and are not direct representations of the oral language (Clackson, 2011, pp. 397-399,434-438). The problem we face here is comparable to the difficulties we face when trying to reconstruct the oral sub-elite Latin language from so-called sub-elite texts (§ 1.2.4), although we don't have the problem of (scribal) formulaic regularization. What we are confident is that the oral, informal written, and formal written languages in Rome between 1st century BC and 3rd century AD form a continuum, which is rightfully known as Classical Latin, under which several genres exist.

1.2.3.2 Canonization and fixing of Latin

After the Classical period, Latin in official writings stopped evolution, which is the destiny of all authorial languages and also happened to Greek (Leonhardt, 2013, pp. 71-72). Indeed, if a language first develops a writing tradition and then undergoes considerable change, usually this will leads to the old literature being forgotten by most (if not nearly all) readers (Leonhardt, 2013, p. 144). The study on what happened after a Classical Latin literary canon had appeared therefore had to rely on less known and more indirect sources, as is briefly discussed in § 1.2.4.

1.2.4 Sub-elite Latin

Vulgar Latin as ancestor of Romance languages Before we discuss Roman-era sources for non-Classical forms of Latin, let us first look at the eventual daughters of Latin: Romance languages. Comparison between Romance languages reveals several interesting commonalities: neutralization of Latin long *ē* and short *i* in their modern reflexes and words commonly used in classical Latin works like *loquor* or *pulcher* being completely missing. With these facts, we conclude that there exists a proto-Romance, which is a kind of Latin but deviates from Classical Latin, in which there is no distinction between *ē* and *i*, and *loquor*, *pulcher*, etc. has been replaced (Herman, 2010, pp. 1-3).

Now we go back to earlier sources. Besides reconstruction based on modern Romance languages, data on non-standard Latin can also be found in "mistakes" or "barbarisms", which are likely reflections of aspects of the oral language which has deviated from the classical works, in texts in late Roman Empire. These "mistakes" seems to be consistent with the aforementioned proto-Romance: for example, neutralization of *ē* and *i* is indeed constantly observed in these texts (Herman, 2010, p. 4).

Non-compliance to Classical Latin standards can be observed even in texts that are supposed to be official and formal, especially after the collapse of Western Roman Empire. In *Chronicel of Fredegar* (mid 7th century), for example, we observe aforementioned Romance-like mixing of vowels and incorrect endings which suggest they had been weakened in the oral language (as in later Romance languages). What is particularly telling is hypercorrections which reveal that the author really couldn't tell some Classical Latin sounds apart but was under the vague impression that some sound distinctions were needed in Classical Latin (Leonhardt, 2013, pp. 158-159).

Unity of Vulgar Latin The next question is whether we can say there is one single Vulgar Latin. Regional variance of Latin can be expected to exist, according our experiences with contemporary languages, and do seem to exist according to Latin authors' accounts (Herman, 2010, p. 116). We are however still confident to say that the regional variance was not of very strong significance, for two reasons.

First, non-literary written texts was largely uniform across the whole Empire (Herman, 2010, p. 117): this is partly due to the existence of many (Classical-like) formulaic texts that can be used everywhere, but when deviations from Classical Latin appeared, they too appeared everywhere in the empire (Clackson and Horrocks, 2011, p. 235), though the propagation may take some time (e.g. see § 6.0.1), and the statistical frequencies of non-standard features (attested ubiquitous) vary regionally (Herman, 2010, p. 118).

Second, Roman grammarians were keen to correct "corrupt" usages of Latin and were known to record local uses, like Africans did not distinguish between long and short vowels. But they reported nothing more fundamental than that: the most notable differences seemed to be pronunciation (Herman, 2010, p. 118).

Based on the available evidence, we can define *Vulgar Latin* as the spoken variety of Latin used by people who were without very formal Classical Latin-oriented school education, which deviates from Classical Latin in a more or less homogeneous way across the Empire.

Elusiveness of Vulgar Latin Although Vulgar Latin, as is mentioned above, is relative uniform, it would be futile to attempt to compose a grammar of Vulgar Latin comparable to a grammar of Classical Latin.

The first reason is the fact that although regional variance in Vulgar Latin is mostly ignorable, the language does have chronological development, and stylist variances in different communities and spheres are to be expected. Conscious and unconscious prescriptive pressures can also cancel existing variances.

The second reason is about the nature of materials we have about Vulgar Latin: as is mentioned above, they were all "Classicalized" when written. Two extreme perceptions of Vulgar Latin are both incorrect. Vulgar Latin is *not* an unattested language that is quite different from Classical Latin and can only be reconstructed by the comparative method, as we have texts. But Vulgar Latin is also not another literary language besides Classical Latin, because these texts are strictly not truly "Vulgar Latin texts", as Romans did not have field linguists to record Vulgar Latin speeches as they were, and all "Vulgar Latin texts" inevitably have both Classical and Vulgar traits (Herman, 2010, pp. 8,26). If a feature that's absent in Classical Latin appears consistently, it has to be a Vulgar, but if certain deviations from Classical Latin is absent, we do not know if it's because it's because Vulgar Latin (in the given time period) was indeed consistent with Classical Latin or if it's because relevant deviations have been regularized by scribes.

This is probably most obviously demonstrated in *Gaius Nouius Enus*, the permission of extension of a loan written on a wax tablet dated to 39 AD. Some consistently appearing spelling "mistakes" clearly show that the spoken language of the author is not precisely the same as Classical Latin and a large portion of the deviations are in accord with spoken Latin revealed by Pompeian graffiti. However, some sub-standard features which are observed in *earlier* periods are absent in the text, likely because they were stigmatized and the author didn't want to sound "rural". Despite the sound changes in the phonology, the text is still pretty comprehensible within the grammatical framework of Classical Latin, which is not surprising given its formulaic nature. Interestingly, some archaisms can be observed in the text, clearly demonstrating that the author was very serious about it, so any spelling mistake has to be because the author did not have access to the "right" pronunciations, not because he had a sloppy attitude. So now what we have is a mixture of some intended archaic spellings, the Classical grammar and a large portion of Classical words, and some sound variances cross validated by other sub-standard texts, while other changes, attested elsewhere, are not present (Clackson and Horrocks, 2011, § 7.4.1).

A final remark is that saying "Romans spoke Vulgar Latin and wrote Classical Latin" still does not accurately reflect the language situation of Roman Empire: besides Latin, we have Greek in the eastern regions, and bilingualism and language mixing are common (Clackson and Horrocks, 2011, § 7.2).

Christian Latin Now we turn to various sources of Vulgar Latin texts. In Christian texts, a tendency is that in the Empire period, early texts reflect *more* spoken features than later texts do: early Christians were often poorly educated, while later Church Fathers had had a good literary education. The same pattern can also be seen in how Neo-Latin replaced Medieval Latin. Many works of the latter however are still Vulgar in the sense that their purpose was to reach the widest audience possible, and definitely some features of spoken syntax and vocabulary were mixed into the texts (Herman, 2010, p. 24).

1.2.5 End of Latin as a living language

Evolution into Romance languages At some point, deviation of Vulgar Latin described in § 1.2.4 finally reached a breaking point such that it's better to stop calling it Vulgar Latin at all: if speakers wanted to learn Classical Latin, they could only learn it as something brand new, not as some prescriptive rules that "purify" their mother tongues ("don't forget to use the ablative here!"), and a speaker without any literary training would be unable to comprehend a Classical Latin article read to them. The problem is when was that point.

It seems that until the 7th century AD, the spoken language was still some sort of Latin: saints' lives were read directly to the congregations in Latin, with the expectation that they could understand them. In Gaul, the first piece of evidence of lack of intelligibility was King Charlemagne's observation that even the clergy could not always understand the meaning of Lord's Prayer. Later we have the famous phrase *lingua romana rustica* mentioned at the council at Tour in 813 (Herman, 2010, p. 114). The Romance part of Strasbourg oaths give us a glance into what *lingua romana rustica* looked like: the text can still be mapped to Classical Latin word-to-word, but the forms of the words had changed so much that intelligibility between the language reflected in the oath and Classical Latin had been greatly weakened (Clackson and Horrocks, 2011, pp. 300-301).

In other areas, evidence for evolution of Latin into Romance was one or two centuries later. And even after Classical Latin became largely unrecognizable to speakers of Romance languages, the memory that the local vernacular and Classical Latin have a shared origin may still be maintained. In Italy, as late as the fifteen century, Latin and Italian were respectively known as *grammatica* and *volgare*, i.e. two varieties of the same language (Leonhardt, 2013, p. 146).

The fate of spoken (Classical-like) Latin The birth of Romance languages does not necessarily mean the death of (Classical-like) Latin as an oral language. In Rome this was preserved at least as late as 1000 AD, as is exemplified by the epitaph of Pope Gregory V, which mentions that he spoke Latin besides a possible Romance language (Clackson and Horrocks, 2011, p. 268).

1.2.6 Coverage of this note

This note is about **Classical Latin** – the Latin of classical Latin writers – and **Ecclesiastical Latin** – the kind of Latin of the Catholic church. That's to say Old Latin, vulgar Latin (with prototypes of Romance articles), etc. are not discussed in detail in this note. Still, some historical knowledge is important for us to understand why Latin is the way it is.

1.3 Sociolinguistic status

1.3.1 Latin before and during early Republic

Etruscan: an early supraregional language The historical and contemporary importance of Latin of course doesn't endorse it as a inherently superior language. Indeed, we only have a handful of Latin texts before 600 BC; as a comparison, there are about 150 pre-600 BC Etruscan texts. Even in the period between 600 BC and 100 BC, during which we have around 3000 Latin texts, we have about 9000 Etruscan texts, which is three times as many as their Latin counterparts. Early Roman elites were partly educated in Etruscan, which lasted to as late as 4th century BC (Leonhardt, 2013, p. 43).

Oscan: a competitor of Latin Oscan was a supraregional language before the rise of Latin, and it seems people speaking the language were recognized by Romans as *Osci*, a concept based not on ethnicity but on language. It's possible that Oscan had a literary tradition (see the discussion below about Greek), which was not preserved in inscriptions, probably for the same reason the great works in Latin were not carved into stones either and that this tradition was possibly semi-oral (Leonhardt, 2013, p. 51).

Greek: the model of literature language The most notable language that was more prestigious than Latin and highly influenced Latin literature was Greek. It is well known that the Roman written culture derived its form from the Greek.

Box 1.2: What type of Greek?

Koine, or Attic, or something else?

Perhaps Romans were not the only people impressed by Greek culture: Greek tragedies were translated into Etruscan and played, and Lucanians – likely Oscan speakers – appropriated Greek art for tomb paintings. The whole Italy was deeply influenced by Greek literature. Indeed, many early Latin poets were not native speakers of Latin: they were multilinguistic and lived on adapting Greek themes and genres and mixing them into Latin. It's more than likely that they would do the same for various Italic languages. Ennius, for example, explicitly said he had three harts, a Greek, an Oscan, and a Latin. Maybe Oscan had its own literature as well, which unfortunately wasn't pass to us because only Oscan inscriptions survived. The influence of Greek culture to Romans was therefore both direct and indirect: some Greek influences were adapted in the whole ancient Italy, and then Romans adopted them (Leonhardt, 2013, pp. 49-50).

1.3.2 Latin in late Republic and early Empire

The rise of Latin seems closely related to the rise of Romans as a politic superpower in Italian Peninsula. The superiority of Greek however had lasted at least to the first century AC, when Latin literature only had its audience in Rome but Greek works was welcome everywhere (Leonhardt, 2013, p. 52).

Box 1.3: When did the status of Greek die away in Western Europe?

When the empire collapsed?

1.3.3 Latin in Medieval times

Before Carolingian Renaissance The prestige of Latin never vanished during the Middle Ages: in former Roman areas, since "barbarians" had long been incorporated into Roman society, the Latin heritage was just accepted as a part of their culture (Leonhardt, 2013, p. 137), and since missionaries played an important role in the culture of non-Roman areas, Latin works were highly regarded because they were essential for the training of future Roman Catholic clergies (Leonhardt, 2013, p. 138).

This doesn't mean authors were always capable to produce Ciceronian Latin works. After the collapse of Western Roman Empire follows a period when truly Classical Latin works ceased to be produced: writings from this age were usually deeply influenced by Vulgar features (§ 1.2.4). It is likely that there was almost no language communities that can use Latin for some sort of communication (Leonhardt, 2013, p. 160).

Carolingian Renaissance

Box 1.4: Carolingian Renaissance

- Existing Latin grammars
- Irish scholars

- Virgil and Cicero, for example, were read solely for the purpose of learning what's good Latin (Leonhardt, 2013, p. 170): can we say the way Carolingian Renaissance revives Latin depended on available Latin texts at that time?

A living second language In the rest of the Middle Ages, the status of Latin was somewhere between a truly dead language and a normal living language (Leonhardt, 2013, p. 145-150). Latin was still live in the sense that people learned Latin the same way modern people learn languages like Spanish or Mandarin, that's to say, they learned Latin by memorizing a limited set of essential grammatical markers (in the case of Mandarin, they are "empty words", and in the case of Latin they are inflectional paradigms) and engaging with an existing community under practical situations: the overwhelmingly rule-based learning style was a much later invention. On the other hand, Latin differed from a true living language because what was acquired by the learner was always a "learner's language", i.e. a possible mental grammar that is *one* explanation of the Latin text that the learner encountered, but probably not *the* explanation that a Roman had in their mind. When learning a true living language, on the other hand, the learner's language is only auxiliary and is always subject to correction by native speakers. Latin therefore was a living second language, a language that was supposed to be living but no one spoke as their mother tongue. The situation is not unlike English outside the core Anglosphere, where people use English, the primary goal is often not to communicate with native speakers (often there is none) (Leonhardt, 2013, pp. 150-154).

Latin's competition with vernaculars In some senses, the status of Latin as *the* true literary language protected – not eliminated – our understanding of early European vernaculars, because if one vernacular – say Frankish – was chosen by Charlemagne as the official language, it soon would undergo the first congress phenomenon aforementioned and then developed a high variety, and the low varieties would then be poorly documented, as in the case of Vulgar Latin (Leonhardt, 2013, pp. 143-144).

Of course, if the status of Latin was so strong, no written materials would be produced in vernaculars at all. The interaction between Latin and the vernaculars therefore is an interesting topic. A general observation is that written vernaculars appeared first in non-Roman areas, and then in formerly Roman areas. This could partially be explained by a residue of the continuum between purely vernacular languages and the literary standard in formerly Roman areas: since Classical Latin could still be partially (although possibly limitedly) comprehended by speakers of local vernaculars, no attempts would be make to write anything using local vernaculars, just like how modern Greeks once rejected translating the Bible into Demotic Greek.

This mechanism however can only explain the *absence* of vernacular literature in formerly Roman areas, but the *presence* of vernacular literature in non-Roman area does not automatically come into being because of lack of Latin capacity: it's definitely possible that these areas were so culturally impoverished that no texts – vernacular or Latin – were published at all. This however is definitely not the case because we actually observe a *positive* correlation between vernacular literature and Latin literature (Leonhardt, 2013, pp. 167-168).

Latin's competition with Greek and Arabic Unlike Greek, Latin had never gained much interest outside Western Europe.

Box 1.5: Latin literature outside Western Europe

- Arabs' attitude
- Byzantine

1.4 Previous studies

1.4.1 Modern linguistic studies

It is not possible to give here a satisfactory overview of the existing literature on Latin morphosyntax, simply because there are too many works devoted to the topic with too long a historical time span. A detailed literature review is essentially a history of linguistics in the West and far exceeds the scope of this note.

Earliest books covering Latin grammar as a topic of (usually diachronic and comparative) interest and academic value on its own (and not as mere preparation for students to read texts) were often named *new* Latin grammars. Among them is a volume produced a century ago and yet no less valuable, Allen and Greenough (1903).

Latinists tend to say that they are influenced by the *functional-typological tradition* since the emergence of modern linguistics. It should be noted that the term often merely means a focus on descriptive studies of concrete languages inspired by observations from other languages (*typological*) and pragmatic factors in grammar (*functional*), and nothing beyond that. The Oxford Latin Syntax (Pinkster 2015; Pinkster 2021) is Harm Pinkster's *magnum opus*, which supposedly is carried out under the framework of Functional Grammar, as indicated in the preface. Yet the Syntax is not formulated in a predication-to-surface form pipeline as found in Simon Dik's works.[1] The main influence of Functional Grammar to Pinkster (2015) and Pinkster (2021) is perhaps the terminology in the Syntax, which nonetheless appears easily translatable to other theoretical frameworks.

Latin has also gained attention from linguists with a more openly confessed structure-oriented perspective of language. Chomskyan studies on Latin morphosyntax include Devine and Stephens (2006), Danckaert (2015), Danckaert (2017); Oniga (2014) and many more (Mateu and Oniga, 2017). One problem of the generative approach or indeed any theoretical linguistic approach is the lack of native speakers' grammatical intuition. In this way the "functional-typological tradition" is often of practical advantages as it does not intend to arrive at all and only acceptable forms of Latin. Corpus linguistic tools are widely used in the functional-typological approach (but is

[1] A slogan found in both the Syntax and reviews of it is that it treats Latin as a system for communication. Structure-based approaches to language however do not deny that language is a tool for communication: what is stated instead is that linguistic structures are based on primitives perhaps not directly reducible to communicative intentions, and the way these primitives are put together is nonetheless obviously highly influenced by communication. On the other hand, a serious communication-based grammar has no choice but to have an articulated theory of the structure of communication intentions, based on which all permissible forms are to be derived, which is nowhere to be found in the Syntax. From our perspective this should be considered an advantage and not a disadvantage. We also note that the prediction-syntactic function-pragmatic function pipeline, as a descriptive tool, has little difference from the vP-TP-CP hierarchy in generative syntax. One can even argue that Pinkster utilizes the notion of an argument hierarchy (in his words, first, second, third arguments).

not limited to this approach; Danckaert 2015).

Thus the functional-typological tradition is better contrasted with all theoretical linguistic approaches, in that the latter attempt to demarcate what is and is not possible for natural languages, while the former relies on the "common wisdom" or "Basic Linguistic Theory" (Dixon, 2009) on the matter. In this way the functional-typological tradition connects well to the traditional philological tradition (and may be known as ethnic philology when the language being studied is previously less well known).

1.4.2 Earlier works

What makes

1.5 Texts

A brief summary of genres of Latin texts has already been provided in § 1.2. Here we elaborate more on resources

Box 1.6: List of Latin texts

- Old Latin texts: archaeology
- Old Latin texts: Roman authors
- The road to standardization
- The Classical canon: Cicero, etc.
- Legal and technical Latin, and Latin historiography
- No one read Classical Latin after Christianization, except in Ireland
- Latin inscriptions, curse tables, etc. in Empire period
- Medieval Latin, Neo Latin

Chapter 2

Phonology and the writing system

2.1 Phonemes and the alphabet

Although the phoneme inventory of a language often is not accurately reflected by its preferred writing system, since Latin is a classical language and no ancient Roman is alive today, its phonology has to be inferred from known texts.

The most accepted writing system of Latin developed into what we call **Latin letters** – or the **Roman alphabet** – today, which is the most widely used writing system in the world. **Old Italic scripts**, used by Early Old Latin inscriptions as well as neighbor languages, show a larger degree of variation, which clearly derived from Greek letters. The standard Latin alphabet derived from old Italic scripts. Note that ancient Romans only used the big letters; the small letters was invented during the era of Charlemange.

The letter *J* was not used by ancient Romans, although we sometimes see *I* appearing at the start of a word and therefore possibly represents the semivowel /j/. The letters *U* and *W* are also not used. Similar to the case of *I*, the letter *V* is used to represent what appears to be the semivowel /w/ as well as the vowel /u/. The letter *K* is an archaic one and only appears before *A* in a small number of words (Oniga, 2014, chap. 2). The letters *Y* and *Z* are used to spell Greek words that include TODO: Y and the voiced dental affricate, respectively.

2.2 Consonant inventory

The Latin consonant inventory is therefore given by Oniga (2014, Table 3.1). Note that the letter *X* is a double consonant: it means /ks/ or /gs/.

2.3 Vowel inventory

Two semivowels – /j/ and /w/ – can be recognized, which appear as *i* or *u*.

The vowels are given by Oniga (2014, Table 3.2). Each vowel has a long variety and a short variety.

Historical evolution of Latin vowels

2.4 Prosody

2.5 Morphophonological rules

Some phonological rules in Latin are sensitive to morpheme boundaries. We can therefore assert that at least in a historical stage, Latin speakers had a clear sense of morphemes as real phonological objects, instead of mere theoretical models.

2.5.1 Vowel deletion

Short vowels *a*, *o* and *e* become zero before a morpheme boundary or another vowel (Oniga, 2014, § 8.3). This rule is exemplified by the absence of the thematic vowel in both declension (TODO: rosis) and conjugation (TODO).

2.5.2 Vowel shortening

A long vowel before another vowel or morpheme boundary is not deleted, but shortened. Again this is exemplified (§ 8.3.1, TODO: ref)

Also, in the final syllable of a phonological word, a long vowel before a consonant except *s* is also shortened. Counterexamples when the vowel is not in the final syllable exist, like *bāris* 'a type of flat-bottomed freighter used on the Nile in Ancient Egypt'.

A long vowel is generally shortened before a sequence containing a liquid or nasal and a following stop consonant, like *nt*. This rule comes from an older Indo-European sound law: the Osthoff's Law (Oniga, 2014, p. 55).

2.5.3 Vowel weakening

When a short syllable is in a medial, open syllable, and a morpheme boundary occurs immediately before, within or after the syllable, it becomes *i* (Oniga 2014, p. 55; TODO: ref).

2.5.4 Vowel lengthening

A vowel is always *lengthened* before *nf* and *ns* (Oniga, 2014, p. 55).

Chapter 3

Grammatical overview

This note takes a approach inspired largely by Distributed Morphology and Cartography, and pays special attention to word formation and its interaction with syntax, structural parallel between morphology and syntax and its absence, and hierarchical structures and their relations with word orders.

In this way, we take a structure-oriented approach and first go over clauses and noun phrases in Latin and make extensive comparison between constructions in Latin and their cross-linguistic counterparts. Parts of speech definitions are then made, again on a cross-linguistic comparative basis.

Traditionally, Latin grammars take a parsing oriented, morphology-to-syntax approach. Discussions on parsing Latin texts are given in § 3.6.

3.1 Morphological typology

3.1.1 Wordhood

Latin is well known for its rich morphology. Although wordhood in Latin is easier to define than it is in some other languages, a discussion may still be beneficial.

A review of traditional criteria of grammatical wordhood can be found in Dixon (2012, pp. 12-19). The list is heterogeneous. Some criteria in the list are not always uncontroversially about wordhood. The criterion of conventional meaning is idiomization and applies to prototypical phrasal idioms too. The criterion of morphological processes risks circular argument, because affixation and compounding do not have clear-cut differences from prototypical phrasal operations without other criteria. Non-concatenative processes are much more likely to be morphological, but this is still not absolute.[1] In general, structure building processes cannot always be decisively classified as morphological or syntactic without consulting other criteria.

Beyond these less specific criteria, wordhood can be relatively uncontroversially tested by criteria in Dixon (2012, pp. 12-19) divided into two groups.

First, wordhood is clearly manifested when roots and grammatical formatives are brought together to form a sequence which is not clearly motivated by anything resembling prototypical phrasal syntax. This covers the criteria of fixed order of morphemes, cohesiveness, inflection,[2] and non-recursion. Some of these criteria are also

[1] Consider, for example, separable verbs in German, which can be analyzed syntactically, and their word-like properties can be attributed to idiomization (Wurmbrand, 2000; Müller, 2002).

[2] Categories marked by inflection can in theory also be marked by auxiliaries, which reflect the

not absolute and are tendencies. These criteria demarcate word boundaries in addition to justifying existence of words.

Grammatical processes happening within word boundaries demarcated by these tests may still resemble syntactic processes, but this does not invalidate their intra-word status, especially when the form in question has phonological wordhood that is otherwise hard to explain. This process seems sporadic in Latin. One clear example is *duo-dē-vīgintī* 'two from twenty (i.e. 18)', which obviously is sealed between word boundaries as it does not decline; one potential example is *senatusconsultum*, which was sometimes written as a single word.

Second, even when word boundaries are not immediately clear, subcategorization, or more generally, syntactic behaviors of a form not predictable from its internal makeup typically indicates wordhood (also see § 3.5.4). This criterion of (syntactic) lexicalization is especially useful for studying languages with highly productive roots, where structure building processes have no clear morphological/syntactic distinctions.

Wordhood in Latin can be almost solely defined via the first group of criteria, because of the clear inflectional endings demarcating the right boundary of a word (e.g. 1). No inconsistency is observed between this and the second group of criteria.

Furthermore, Latin grammatical wordhood agrees with phonological wordhood almost without exception, the latter definable via stress patterns. The traditional wordhood definition, expressed by orthographical conventions, largely agrees with both grammatical and phonological wordhood and is therefore linguistically well understood.

In this way, establishing wordhood of words without inflectional morphology (traditionally known as *particles*) based largely on phonological wordhood should not be controversial.

Subtleties in wordhood definition mainly revolve around clitics, the most important one being the coordination *-que*: they are not a part of inflectional morphology of any stem but have to be attached after another word. The orthographical standard simply dictates that they are to be recognized as a part of the words they follow (TODO: stress?).

3.1.2 Concatenative morphology

A Latin content word that inflects can be uniformly divided into a stem and a suffixal inflectional ending, enabling a clear derivation-inflection distinction (1). This distinction then defines terms like *lexeme*, *stem* and *ending*.

This distinction is based on purely morphological considerations: the formation of perfect stems, despite marking an aspectual category, clearly falls in the derivational part of verbal morphology (§ 8.1.1). Contextual allomorphy accompanies concatenative morphology (§ 2.5).

Despite its richness, a large portion of instances of Latin derivation are historical, with meanings of derived forms having significantly shifted and no longer regularly inferrable.

Prefixes exist in stem formation only, while suffixes appear in both derivation and inflection. It appears that grammatical information about how a form interacts with

hierarchical syntactic structure of the structure containing the word much more transparently. That the inflection markers somehow have to attach to the head word of a construction on the other hand cannot be motivated by purely compositional reasons, and thus indicates wordhood.

its environment (change of part of speech, case, valency, etc.) is almost always carried by suffixes.

Compounding exists in Latin does use compounding (1): two roots may be joined together by a connecting vowel (Smith 2016, § 92). Compounding however was less productive – if not obsolete – in Classical Latin.

(1) $[[\text{aequ-i-libr}]_{\text{stem}}\text{-}[\text{ium}]_{\text{ending}}]_{\text{grammatical word}}$
even-ATTR-balance-SG.NOM
'equilibrium'

3.1.3 Non-concatenative morphology

Concatenative morphology (affixation and compounding) is prominent but isn't the only morphological device.

First, it should be noted that Latin concatenative morphology involves stem alternation. (2) illustrates a small part of the Latin verb paradigm (where we do not perform a fully compositional analysis of the endings; TODO), in which we observe that the verb stem has three different forms in the three inflected forms.

(2) a. am-ō
love-PRS.IND.1SG
'I love'

b. amā-s
love-PRS.IND.2SG
'you love'

c. ama-t
love-PRS.IND.3SG
'he/she/it loves'

Reduplication is attested in formation of the perfect stem (TODO: ref); this however is largely historical and is no longer productive.

Dropping of first-conjugation stem-final vowel (§ 8.3.1) may be analyzed as subtraction, although it can be seen as due to morphophonological rules. The imperfect *-ba-* is sometimes said to be an infix (as well as its counterparts like *-bi-*), though it fits in a concatenative picture of verbal morphology.

Residues of the ablaut system in PIE can be observed in Latin. In the third declension, for example, we have the following alternation: *gen-us* 'family-NOM.SG'/ *gen-er-is* 'family-GEN.SG', and since *-us* is *-os* in earlier stages of the language, we observe an alternation between *o* and *e*. Another famous instance is the alternation between *toga* and *tegō* 'cover'. It's also possible that no vowel exists at the place where we find *o* and *e*. We therefore find a vowel gradation system with three variants: ∅, *o*, and *e*. Note that what controls the vowel gradation system is grammatical categories like case, etc., and therefore ablaut is a part of morphology, not just phonology (Weiss, 2009, p. 46).

3.2 Clausal syntax

3.2.1 Clause types

A clause may appear as a single utterance called a **sentence**,[3] which is either declarative or interrogative. A clause can also be embedded (§ 3.2.2.3).

Regarding the internal structures, Latin clauses types are traditionally grouped into finite and non-finite ones based on verbal morphology and subject behaviors.

Sentential clauses are always finite. In sentential clauses, the category of speech force is not marked by verb morphology, and instead is marked by e.g. fronting of the interrogative pronoun (3). Latin lacks grammatical markers (e.g. sentence-final particles) appearing only in sentential clauses and not embedded finite clauses.

(3)	**Quid**$_{\text{WH},i}$	**enim**$_{\text{CONJ}}$	dici-t	[aliud]$_{\text{object},i}$	[C.	Numitori-us]$_{\text{subject}}$?
	what-ACC.N.SG	indeed	say.PRES-PRES.IND.3SG	other-ACC.N.SG	Gaius-NOM.SG	Numitorius-NOM.SG

‘Then what else does Gaius Numitorius say?’ (Verr. 2.5)

Non-finite constructions include INFINITIVES and PARTICIPLES, and the GERUND. These constructions can be distinguished by morphology only: the head verb of a infinitive clause in Latin has an independent cell in its inflectional paradigm. This is to be contrasted with the case in modern English, where infinitive clauses can be recognized based on syntactic criteria, but not morphological forms of the verb (Huddleston and Pullum, 2002, p. 89).

Particles describing the role of a clause in the conversational context (‘indeed’, ‘but’, etc.) exist in both finite (3, where *enim* is a postpositive particle) and nonfinite constructions. These particles are however not coordination or subordination markers.

3.2.2 Internal structure of clauses

All types of clauses share a similar internal makeup. (4) is an example of a Latin clause.

A clause contains a (omissible) subject (*Caesar*; § 3.2.3), a main verb (*fecit* and *profectus*, coordinated) and a possible auxiliary verb for periphrastic conjugation (*est*; chap. 8), tense, aspect, mood (TAM) adverbs (*statim*; § 3.2.4), and core and peripheral arguments (the agent, patient, time, goal phrases; § 3.2.5). The status of predicate is discussed in § 3.2.2.1.

The main verb agrees with the subject (3SG here). The inflectional ending of the main verb, together with TAM adverbs and the possible auxiliary verb, marks TAM grammatical categories (§ 3.2.4).

The argument/adjunct distinction, also known as the complement/adjunct distinction and core/peripheral distinction in the literature, is discussed in § 3.2.5.1.[4]

[3] Some authors (including some Latinists) use the term *sentence* as a synonym of *utterance*; it is not followed in this note.

[4] In some works on English syntax, e.g. Huddleston and Pullum (2002), the term *complement* means a dependent position which, according to various standards, is more closely related to the lexical head; in the case of clauses, a complement is just an argument.

In traditional Latin grammar, however, *complement* means the copular complement. This note fol-

Latin clauses also have abundant subordination (e.g. the temporal clause in 4) and coordination devices (e.g. the branches connected by *et*) (§ 3.2.2.3).

(4) [Post ann-um]$_{\text{time}}$ [Caesar]$_{\text{subject, agent}}$ [[Roma-m]$_{\text{target}}$
after year-ACC.SG Caesar-NOM.SG Roma-ACC.SG
regress-us]$_{\text{time: participle clause}}$ [[quarto]$_{\text{iteration}}$ [se]$_{\text{theme}}$
regress.SUP-NOM.M.SG fourth-ABL.SG REFL.ACC.M.SG
[consule-m]$_{\text{object copular complement}}$ fec-it et
consul-ACC.SG make.PERF-PRF.IND.3SG and
[statim]$_{\text{proximative}}$ [ad Hispanias]$_{\text{goal}}$ est profectus]$_{\text{predicate}}$.
immediately to Spain-ACC.SG COP.PRF.IND.3SG set.out.PERF-NOM.M.SG
'After a year, Caesar, having returned to Rome, made himself consul for the fourth time, and immediately set out for the Spain territories.' (Eutr. 6.24)

3.2.2.1 The question of configurationality

Unlike some other languages where information structure-induced word order changes is limited to sentential clauses, both sentential and embedded clauses in Latin have rather variable constituent orders.

The flexible constituent orders of Latin have led many to characterize it as a free-order or non-configurational language. Frequently cited evidence, like word order permutation or discontinuous constituents, does not decisively determine Latin's non-configurationality, as these phenomena can still be understood in a movement-based account (Danckaert 2017, pp. 73-75; Devine and Stephens 2006, pp. 524-611). The notion of non-configurationality itself has been debated extensively. Recent studies suggest that configurational phenomena can be observed even in languages traditionally considered prototypically non-configurational (Niedzielski, 2017; Morris, 2018; Legate, 2002, among others). In particular, doubts have been cast on the existence of alleged alternative clausal structural building mechanisms like the pronominal argument hypothesis (Legate, 2002).

Given Latin's status as a dormant language, exploring its configurational status is valuable but challenging. Most existing works therefore largely approach the topic in a configurationality-neutral way (Danckaert, 2017, pp. 9-12), tentatively surveying the correlation between the surface word order and various factors (Pinkster, 2021, pp. 948-967). Nonetheless, from the existing texts, features that are otherwise hard to explain without configurationality have been identified.

First, as a bare minimum, asymmetric dependency relations exist in Latin. This is illustrated by e.g. the distribution of the reflexive *se*, which does not appear as the subject. The phenomenon can be interpreted as other arguments being inside the scope of the subject. Such asymmetric phenomena can at least in theory be captured by a constituency grammar with an elaborate hierarchy of pre- and post-nucleus clause positions which constituents can be moved to, although this does not meet the usual standard of configurationality.

lows the terminology used in most descriptive grammars, so use the term *copular complement* to refer to the *predicative complement* in Huddleston and Pullum (2002).

A further confusion, as is seen in § 9.1.2, is due to the term *complement-taking verb*, i.e. a verb that take a complement clause or something semantically equivalent to a complement clause as one of its arguments.

Second, configurational traits can be observed in Latin at the periphery of clauses. This is illustrated by the distribution of the interrogative pronoun *qui*: in proses, it never appears after the main verb of the clause, and when contents of the relative clause are lifted before *qui*, a topic before focus order is usually expected (Danckaert, 2017, pp. 19-20). One can thus analyze the left periphery of the Latin relative clause as a rigidly positioned relative pronoun with a hierarchy of topic and focus positions above it.

Third, the relative order between the negator and the auxiliary strongly suggests configurationality. It is rare, if not impossible, for *non* used as a clausal negator (and not in e.g. *non ... sed ...*) to appear before the highest verb/auxiliary in a clause (Danckaert 2017, pp. 39-44; Devine and Stephens 2006, p. 183). This is consistent with the following analysis: the negator precedes the highest verb/auxiliary in a immobile nucleus clause; lower constituents and the negator can move out of the nucleus clause to the left, but a locality constraint prohibits the highest verb/auxiliary from passing the negator (Danckaert, 2017, pp. 60-62).

Fourth, it appears that at least in some examples, standard constituency tests based on coordination (4), fronting, ellipsis, etc. suggest existence of a "verb phrase" or "predicate" (Danckaert, 2017, pp. 68-73).

Finally, corpus analyses of Latin reveal that the frequencies of the order between the verb and the object appears to be conditioned by whether there is an auxiliary verb or a modality verb in the clause, which is analyzable in a configurational framework (Danckaert, 2017, pp. 293-295).

Configurational features thus can be observed in all levels of Latin clausal syntax.

3.2.2.2 Information structure

The main difference between Latin (which now should be more accurately known as a discourse-configurational language) and languages with more rigid word order therefore can be attributed to the former having far more information structural devices (Danckaert, 2017, p. 15).

First, in broad focus sentences in which the whole situation of affairs described by the sentence is the focus (and hence sentence-level topicalization and focalization should be minimal, compared with the case in other constructions; Devine and Stephens 2006, p. 36), permutation of constituent order is still observable, which appears to be explainable as topic-in-focus and focus-in-focus information structure marking at the center of the verb phrase.

Second, cross-linguistically, adverbs' linear order is consistent with their relative scopes. Assuming this to be at least partly true (considering possible movements of adverbs) in Latin, it appears that in broad focus sentences, arguments may precede or antecede adverbs. This means that Latin has both core verb phrase-level information structure movements (Devine and Stephens, 2006, pp. 36-88) and what is known in other languages as scrambling, which lifts arguments to the region of adverbs with scopes over core and many peripheral arguments and has weak topicalization (presupposition) or focalization effects (Devine and Stephens, 2006, pp. 98-112).

Third, sentence-level topicalization and focalization exist in Latin. One example is that it is possible (not necessary) that so-called strong focus arguments (i.e. arguments in sentences that respond to questions like 'who did it' or 'who he did it to'; in Latin this can be recognized in e.g. *x non y* constructions) are raised to positions before the subject (Devine and Stephens, 2006, pp. 225-232,302).

3.2.2.3 Clause combining

So-called clause combining is sometimes taken to include serial verb constructions and other low-level coordination devices in the clause. In Latin however there is no serial verb construction.

Subordination[5] constructions include complement clause constructions (a clause being a part of the argument structure of another clause), relative clause constructions (a clause being a dependent, usually a high-level modifier, in the nominal system) and adverbial clause constructions (a clause being high-level modifier in the verbal system specifying e.g. the condition or the consequence of the main clause (subordination)).

Coordination in Latin is marked by *et* or the clitic *=que*.

3.2.3 Subject

The distribution of the nominative case implies that Latin is a nominative-accusative language. We still ideally desire syntactic evidence to corroborate this conclusion, which requires Latin to have a well-defined clause-level pivot (the subject) that can be identified with the most prominent argument in the argument structure, from a mixed-pivot approach of defining subjecthood (Aldridge, 2008).

Tentatively considering the subject to be the constituent bearing the nominative case, its status as the clause-level pivot can be proven from the following facts.

First, that nonfinite clauses in Latin are either subjectless or have have a non-nominative clausal dependent resembling the finite clause subject. It is common for deficient clauses to have no canonical subject, cross-linguistically.

Second, agreement is often related to subjecthood, though it may also be influenced by factors other than structural prominence of an argument, like discourse participant status (Oxford, 2017). This subtlety does not exist in Latin and the verb always agrees with the subject in the number and person features, which can also be taken as evidence of structural prominence of the nominative argument.

The two observations above guarantees Latin is not morphologically ergative.

Third, in Latin, in two coordinated clauses sharing a subject, the subject usually appears at the beginning of the coordination construction (Pinkster, 2021, § 19.5). This suggests that the subject is somehow structurally promoted to a higher position in this kind of coordination,[6] and is corroborated by verb phrase constituency test (§ 3.2.2.1).

A fourth possible diagnostics for subjecthood is scope, as in interpretation of sentences like *every person carries a gift*. This scope effect however is apparently not well studied in Latin.

Fifth, in clauses with neutral information structure, the subject often appears at the beginning of the clause (Devine and Stephens, 2006, p. 79). As said, though, information structure easily permutes the word order (§ 3.2.2.1).

A subject in Latin is an noun phrase or a complement clause, without known structural complexity constraints. It however cannot be a GERUND (§ 8.5.2.1).

[5] The term *subordination* sometimes is used as a synonym of *clause combining*. Here we use *subordination* in a narrower meaning.

[6] Omission of the object in coordination, on the other hand, shows more diversity in constituent orders (Pinkster, 2021, § 19.6), suggesting that the exact structural analysis may also be more complicated.

In traditional grammar, the non-subject parts of a clause (§ 3.2.2) are collectively known as the predicate (Allen and Greenough, 1903, p. 164). The predicate (known also in other languages as the verb phrase) appears to have constituent status (§ 3.2.2.1), although multiple topicalization and focalization render the concept with limited value in surface-orientated analysis.

In Latin we can also recognize an argument structure-level pivot, i.e. the A argument (§ 3.2.5.3). The fact that in active clauses, the argument structure pivot is always the clausal pivot i.e. the subject means Latin is indeed a nominative-accusative language.

The subject position can be left empty for pragmatic reasons, and it is also possible to have an empty, indefinite subject (Pinkster, 2015, § 9.9).

3.2.4 Tense, aspect, modality, and all that

Tense, aspect and modality categories may be marked by verbal inflection (chap. 8) or by adverbs, whose linear order, cross-linguistically, tends to agree with their relative scope (Cinque, 1999). We list some of the obviously attestable categories below.

These categories can (cross-linguistically) be roughly divided into non-aspectual categories with wider scope, which include speaker-oriented categories, modality and tense, and aspectual categories with narrower scope.

3.2.4.1 Speaker-oriented categories

Latin obviously has grammatical speech forces, namely the declarative, the interrogative (shown in fronting of WH-words), and the imperative (reflected by verbal morphology). Latin does not have uncontroversial marking of evidentiality but there are hypotheses relating it with the alternation between *Accusativus cum Participio* and *Accusativus cum Infinitivo* (Greco, 2013).

3.2.4.2 Tense and modality

Epistemic modality is marked morphologically by the MOOD in declarative and interrogative clauses (Pinkster, 2015, p. 388). Epistemic modality is additionally marked by the adverb *fortasse*, whose linear order is consistent with its TAM status (Danckaert, 2017, p. 27).

The primary tense category establishes one (or two) *reference time* (see discussions on the anteriority category below) and compares its relation with the *speech time* (Cinque 1999, pp. 81-83; Huddleston and Pullum 2002, pp. 126-127). Latin has a past/present/future tense system shown by the conjugation endings. The past future tense seen in English (*he [would] take part in in if he knew it*) is absent in Latin.

The morphological MOOD in the imperative construction seems to mark the denontic modality: there is no epistemic modality to mark anyway (Pinkster, 2015, p. 388).

3.2.4.3 Aspectual categories

In Latin, just like in English, the habitual aspectuality is not marked by verbal morphology, but often the PRESENT category has a habitual reading.

Anteriority (Cinque, 1999, pp. 81-83), or secondary tense (Huddleston and Pullum, 2002, pp. 139-140) is about whether the relation between the event time and the reference time (whose relation with the speech time is determined by primary tense). A clear perfect/simultaneous distinction exists in Latin, marked by the alternation between the PERFECT stem and the IMPERFECT stem and also different endings for the two system.

It seems we also have a posteriority value in Latin, because even when the primary tense is absent (as in the two FUTURE participles) or marked otherwise (as in two periphrastic verb forms), a meaning of "future" – not necessarily defined according to the speech time – is still available, and the FUTURE participles rarely appear with *sum* in PERFECT (which encodes anteriority) forms (Pinkster, 2015, Table 7.3).

The point of view aspect compares the position of the event time (which is compared with the reference time in anteriority) within the whole situation: the situation may be described as a whole (*perfective*), or the clause only describes a fraction of a continuous event (*imperfective* or *progressive*).

In Latin we do not have a clear marker of the perfective/imperfective distinction. It is however generally believed that at least the IMPERFECT verb form regularly has an imperfective meaning (Pinkster, 2015, p. 380).

The relation between the point of view and the situation can be more complicated: for example we have terminative, inchoative or retrospective aspects. The Latin *-sc-* infix is an inchoative marker.

The point of view aspect is to be distinguished from the lexical aspect, as illustrated by the English example *I'm closing the door* (imperfective point of view, instant event).

We note that crosslinguistically, habituality and anteriority and point of view aspectuality (see below) can co-exist, as in English examples like *what we feel has [always]$_{habitual}$ [already]$_{anteriority}$ been felt, again and again* or *this life has [always]$_{habitual}$ [already]$_{anteriority}$ [been happening]$_{progressive}$*. The scopes of the three categories follow the order of habituality > anteriority > point of view aspectuality. The combinatorial possibility of TAM markers in Latin appears to be less well studied.

The list above is obviously far from exhaustive. A comprehensive study of Latin's TAM adverbs and their word order is currently lacking. The exact TAM values of Latin verb morphology are also not completely settled, and discoveries about them are still being made (Dyck, 2017).

3.2.5 Arguments

3.2.5.1 The core/peripheral distinction

The argument/adjunct distinction is not always clear in languages. For this reason, in this note, we use the term *peripheral arguments* as in Dixon (2009) to refer to non-TAM adjuncts. Criteria traditionally used to establish the distinction between core and peripheral arguments have been extensively reviewed in McInnerney (2022) and are classified and applied to Latin below.

First, one may use semantic criteria to make the distinction. This is not viable because (a) in compositional semantics no such distinction can be reliably made (McInnerney, 2022, pp. 24-27), and (b) it is cross-linguistically observed that there exists oblique arguments, i.e. instruments, manners, locations, etc. expressions which we have reasons to classify as core arguments for other reasons.

Second, the core/peripheral distinction may be taken literally, if it is demonstrated that some arguments (e.g. the circumstantial phrase) resemble modifiers of the argument structure formed by other arguments (e.g. the agent, the patient, etc.) (Cinque 1999, p. 29).

Under this overall understanding of the core/peripheral distinction, multiple diagnostics exist, including co-occurrence of the verb and its core arguments, replacement of the verb and its core arguments only by a pro-form, extraction constraints, and more (McInnerney, 2022, pp. 106-108). However, whether any *qualitative* difference can be established is highly questionable (McInnerney, 2022, pp. 128,212).

We therefore pay attention to the structural difference between arguments but do not use it as the decisive criterion of the core/peripheral distinction.

Third, the core/peripheral distinction can be understood as lexicalization of arguments: if presence of an argument is dictated by the lexical, unpredictable properties of the verb, then it is a core argument.

The obligatoriness diagnostics is the most straightforward application of this. Caveats include (a) dropping an obligatory argument by various grammatical processes is possible (but typically with e.g. valency alternation or other observable phenomena, like the omitted argument having to refer to a known entity), (b) sometimes but not always, obligatoriness originates from pragmatic reasons (McInnerney, 2022, pp. 113-115), and (c) sometimes arguments may be introduced by idiomization (cf. English *care for* vs. *care about*), which makes it necessary for an argument (often with a preposition) to exist (McInnerney, 2022, pp. 112-113). The core/peripheral distinction is not a necessary concept for these cases above. However, cases in which lexicalization takes place beyond any reasonable doubt exist. No pragmatic or idiomization motivation explains why *slog* in English has to have a locative argument, for instance (McInnerney, 2022, pp. 115-117).

We emphasize that this criterion of lexicalization does not entail any necessary structural differences between core and peripheral arguments McInnerney 2022, p. 125.

In Latin, arguments can be dropped only when pragmatics allows it, which can be interpreted as obligatoriness of arguments. Further, although the relation between semantic roles of arguments and their morphological cases is largely transparent (commonly known as "syntax of cases" in the Latin grammatical tradition), this is not absolute (*persuadere* vs. *monere*), which can also be taken as evidence for syntactic lexicalization i.e. subcategorization of argument structure. As

3.2.5.2 Some "peripheral argument" slots

Some argument slots are conventionally known as peripheral ones for having wider scope (see the second possible criterion in § 3.2.5.1) and may be a part of a verb's verb frame too (see below).

Circumstantial phrases modify the event as a whole and specify when and where it happened. They often are the most peripheral constituents in the (extended) argument structure (Cinque, 1999, p. 29). Manner phrases describe *how* something happened. (Devine and Stephens, 2006, p. 71-75, pp. 101-109).

Instrument phrases are sometimes considered arguments and sometimes adjuncts, which is also the case in Latin (Devine and Stephens, 2006, pp. 57-58, p. 65-67). Crosslinguistically, it seems some instrument phrases are a part of the core argument structure (Pascual Pou, 2001), and some instrument phrases can be regularly promoted to the subject position under proper pragmatic environments (Alexiadou and Schäfer, 2008), in which certain phenomena suggest heterogeneous nature of instrument phrases (Wilson et al., 2020).

3.2.5.3 Core arguments, grammatical functions and voice

We now turn to what are traditionally considered core arguments according to structural criteria (the second in § 3.2.5.1).

Cross-linguistically, in discussing these arguments, it is wise to distinguish between grammatical functions (subject, object, etc.) and argument slots. Argument slots are defined by properties like how reflexives work, semantics of the roles,[7] the semantics difference in valency alternations (§ 3.2.5.4), etc. Grammatical relations are defined by argument extraction (in relativization or topicalization), case assignment, etc. The non-trivial mapping between them is essentially the alignment type of the language and obviously involves certain so-called peripheral arguments too (as in instrument-to-subject promotion).

To avoid confusion with semantic roles, in typology, argument slots are labeled as S, A, P, T, G, etc., to be discussed in next sections. We should caution that there is no guarantee that argument slots traditionally labeled by the same label are of comparable nature: indeed it is well known that S arguments split into A-like ones and P-like ones.

Subject is a well established grammatical function in Latin (§ 3.2.3; Pinkster 2015, p. 28). It is typically the A-argument. The A argument, i.e. the most syntactically agent-like argument or the pivot of the argument structure (Aldridge, 2008), can be distinguished by the fact that (a) it can never be filled by the reflexive *se*, and (b) it is the argument omitted in imperative constructions, and (c) it is the shared argument in control constructions (§ 3.2.5.9).

It is not clear if the direct object is a single grammatical function. Uncontroversial objects are the most prominent non-subject argument that bears the accusative case and can be passivized (indicating absence of inherent case assignment; § 3.3.1; Pinkster 2015, pp. 28-29). This does not at all guarantee a shared structural position: a much more convincing piece of evidence is e.g. a relative clause construction that indicates the grammatical function of the extracted argument by verb morphology (Jacques 2016). This is also a problem in English, in which a notion of (TODO: CGEL).

Similarly, what is an *indirect object* is not clear.

Languages often have valency increasing or decreasing devices, and other alternations like the dative alternation, the locative alternation, the subject-instrument alternation, and the like. The primary valency decreasing device in Latin is the passive voice, in which the A argument in an existing argument structure is suppressed or becomes an instrument-like argument introduced by *ab*, and the second most prominent argument is promoted to the A position.

In Latin, the passive morphological marking of the verb also generalizes to all verbs without a prototypical A. This happens to unaccusative verbs (§ 3.2.5.4), and also in so-called impersonal constructions, where the A is suppressed but nothing is promoted

[7] How uniform the relation between semantic roles and syntactic argument slots is still disputed; see discussions on the Uniformity of Theta Assignment Hypothesis. We should note that two arguments traditionally denoted by the same semantic role label (e.g. *stimulus*) are not necessarily semantically equivalent or syntactically comparable. In Dixon (2005, § 4.2), for example, what bears the label *target* is almost prototypically patient-like in verb frame I and is the goal of a directional construction in verb frame II, and of course there are syntactic as well as semantic differences between the two frames.

to the subject position.

3.2.5.4 Intransitive and monotransitive constructions

A prototypical transitive clause has a A argument and a P argument. The sole argument of an intransitive clause, i.e. the S argument, can be A-like (*[Simon] jumped off*, initiating an event, "*unergative*") or P-like (*Simon fell*, undergoing an event, "*unaccusative*").[8]

In Latin often the transitive construction can be detransitivized by suppressing the P argument (which can possibly be understood as filling the P slot by an empty indefinite pronoun), known as the *absolute use* of the verb (Pinkster, 2015, § 9.16). This leads to a S=A valency alternation. This is also observed in English: *the dogs bited me* v.s. *the dog bites*.

In some transitive clauses, the so-called P on its own already forms a complete (stative) situation of affair, which is caused by an external argument (which is then considered A-like). It's possible that the external argument can be removed, leading to S=P valency alternation, known as *decausative* in Pinkster (2015, p. 281).

In Latin, it seems that unaccusative intransitive verbs are all deponent (Oniga, 2014, pp. 308-309). The passive voice therefore is used whenever an agent is absent. unergative intransitive verbs do not have PERFECT PASSIVE PARTICIPLES, because the passive voice is not compatible with them (Giusti and Iovino, 2019).

The grammatical coding of semantics roles of a verb may vary a lot when the arguments are not semantically clearly agent-like or patient-like (§ 9.9).

The P argument of some Latin verbs does not have accusative case, which appears lexically encoded and has no clear synchronic explanation (cf. English *accuse of*).

3.2.5.5 Dative, benefactive and directional constructions

In a dative construction we have A, G and T arguments: something (T, theme-like) is given to someone (G, goal-like). The meaning and structure of dative constructions is often parallel to that of benefactive and directional constructions.

Structural analysis of these constructions (and hence how homogeneous the properties of G- and T-arguments are) is controversial. It has been proposed that a *small clause* in which arguments are licensed exists in at least some constructions (Wurmbrand, 2000; Punske, 2013), an analysis not agreed upon in all such constructions (Huang, 2007; Punske, 2013). The issue is further complicated by the appearance of particles and/or preverbs, some of which appear to originate from the small clause, while others are likely not (Farrell, 2005; Punske, 2013; Wurmbrand, 2000).

Latin appears to lack verb-particle constructions with movable particles. Some verbs with a preverb ("non-movable particle") highly likely have a directional small clause in their argument structures: the preverb always describes the path, the case marking of the argument(s) are sometimes never attested without the preverb, and the meaning of the verb may also vary depending on whether the preposition is attached to the verb, indicating that it's the preverb that licenses the argument (Mare, 2018).

Many of these verbs govern the accusative case even when the preposition corresponding to the preverb does not, and passivization is possible (Pinkster, 2015, § 4.22).

[8] The unergative/unaccusative terminology is misleading because it describes argument structure phenomena and not alignment.

This seems comparable to certain English preposition verbs where the preposition may be somehow defunct and the location argument it governs syntactically is the object of the verb, while in the argument structure of others, the prepositional phrase is sealed and nothing can be extracted from it (Richards, 2017).

In the argument structure of other verbs with preverbs, the location argument is dative, and can sometimes be understood as the goal of the action (Pinkster, 2015, § 4.25, § 12.7). This, if analyzed synchronically, can be explained by assuming that Latin has certain small clauses that license dative arguments but cannot be standalone prepositional phrase.

It's possible to keep only the dative recipient in the benefactive construction: this leads to two-place verbs governing a dative argument (Pinkster, 2015, § 4.24). We note that the benefactive construction differs from the prototypical dative construction of giving, and the latter is never attested in two-place verb frames (Pinkster, 2015, § 12.7).

3.2.5.6 Comitatives

The English comitative construction has been analyzed in the small clause approach, in which the pivot of the small clause becomes the subject or the object of the main clause, and the rest of the construction becomes an oblique argument (Zhang, 2007). A comitative construction is attested in Latin, with the comitative argument marked by the preposition *cum* (Pinkster, 2015, § 4.38). Structural analysis of this construction appears missing.

3.2.5.7 Copular complements

A copular complement in the argument structure (e.g. *smart* in *I consider Mary smart*) can be viewed as a displaced attributive or appositive (and hence is prototypically filled by a noun phrase or an adjective phrase). This view essentially assumes a small clause structure (Bowers, 1993) but is not without controversies.

In the grammatical tradition of Latin, copular complements are known as *predicates*. Latin predicates agree with the arguments they modify: thus the nominative predicate gives a property of the subject and agrees with it, and the accusative predicate gives a property of the direct object and agrees with it. In passivization of the direct object, the accusative predicate becomes the nominative predicate.

> **Box 3.1: Oblique copular complements**
>
> Other types of copular complements without agreeing with the subject exist. TODO: ablative of quality, price, etc. The syntactic status of copular complements here are closer to PPs: we may say they receive *inherent cases*, while the nominative and accusative copular complements receive *structural cases* (chap. 6).

3.2.5.8 Indefinite objects

Indefinite, non-referential, often idiomized objects are typically seen in Sinitic languages (as in Mandarin *chī fàn* 'eat (lit. eat meal)'; Li and Shi 2003). Non-referential objects also appear in Latin (Devine and Stephens, 2006, p. 54).

A cognate object is a clausal dependent that is obligatorily selected by the main verb and usually overlaps semantically with the content of the main verb. In English we have *Ben died [a long and painful death]* and *Joe smiled [a smile]$_i$ and it$_i$ is a creepy one.*

3.2.5.9 Control and raising

In a control construction, the A argument of a nonfinite complement clause is also an argument in the main clause. In a raising construction, the A argument of a nonfinite complement clause has a grammatical function in the main clause (while not bearing any semantic role in the main clause). Whether the two are related (and whether some so-called control constructions are truly syntactic) has been a disputed topic.

Both constructions are attested in Latin, respectively, as the simple infinitive (Oniga, 2014, pp. 295-298) and as the *nominativus cum infinitivo* (Oniga, 2014, pp. 298-301).

It should be noted that the *accusativus cum infinitivo* construction is comparable to English *for sb. to do sth.*, and is not raising or control; the absence of an explicit complementizer is observed in the *ut* clause as well (Oniga, 2014, pp. 290-292).

3.2.6 Derivation

The term *derivation* covers rather heterogeneous phenomena.

First, some so-called derivation devices target strictly materials within the word, including compounding (Di Sciullo, 2005; Scher and Nobrega, 2014; Siddiqi, 2009) and some affixation processes. The only fundamental difference between these processes and prototypical phrasal syntax is wordhood, defined according to morphology and syntactic lexicalization (§ 3.1.1), though this is not absolute either.[9]

This type of verb derivation is frequent in Sinitic languages, as in Mandarin *kàn-jiàn* 'see (lit. look-see)' and *guān-xīn* 'care about (lit. close/connect heart)'. It however is very rare (if possible) in Latin.

Second, some processes considered derivational involve constituents outside the verb word but attach morphological markers (not auxiliaries or particles) to the verb word: a good example is valency alternation manifested by verb affixation. A clear, cross-linguistic distinction between inflection and derivation of this type is not possible: indeed voice is sometimes considered derivation (Jacques, 2021) and sometimes inflection.

In Latin passivization is considered inflectional, based on considerations of the morphological template of the verb. Latin preverbs, arguably introduced by small clause constructions in some cases (§ 3.2.5.5), is however a good instance of this type of verb derivation.

Apart from preverbs, Latin has the similative suffix *-izā-*

3.3 The noun phrase

3.3.1 Distributions of noun phrases

Noun phrases appear as core and peripheral arguments in clauses, and sometimes as standalone utterances (when, for example, responding to a question).

[9]Consider e.g. English *pre- and post-processing*, in which prefixes are coordinated using prototypical phrasal syntax.

In Latin, the grammatical status of a noun phrase in a syntactic environment is marked by its morphological case and sometimes by the preposition it has.

Cases can in general be divided into structural cases (decided purely by the syntactic environment) and inherent cases (preposition-like, with stable semantics on their own).

Latin is morphologically nominative-accusative and has the two cases as the structural cases in the clause environment. In the noun phrase, the genitive case is structural in that not all uses of the genitive case is explainable in terms of possession (Oniga 2014, p. 244; cf. English *of* in *a man of great talent* and *it's so kind of you*). What is considered as prototypical possession also has strong cross-linguistic variation (Dixon, 2010, pp. 262-263).

Examples of inherent cases include the follows. In directional constructions, the ablative case marks the source, while the accusative case (in its inherent case usage) marks the *goal* (Oniga, 2014, p. 238). The dative case also marks the goal, but this is only possible when the path is marked as a preverb. In (semantically more static) locational constructions, the location argument is always ablative. In clauses, the instrument or similar arguments is marked by the ablative.

Latin prepositions sometimes should be regarded as function words and are comparable to morphological case. The preposition *cum* for example marks the comitative argument structure (§ 3.2.5.6, § 7.4.1). The preposition *ab* marks the agent in a passive clause. It seems we have a clear division of labor in directional constructions in Latin: morphological cases are for marking the type of the location, while prepositions are for marking the path (§ 7.4.1), and hierarchies like this are typically evidence for grammaticalization.

3.3.2 Internal makeup of noun phrases

In grammars of languages with more rigid constituent orders, like Huddleston and Pullum (2002), the noun phrase can then be divided into determiners (§ 3.3.3) and the inner parts (the nucleus noun phrase).[10] Similarly, a clause is a nucleus clause plus information packaging and/or marking of clause types and the grammatical function of the clause (§ 3.2.2).

The nucleus noun phrase can further be divided into an argument structure surrounding the head noun (§ 3.3.6), and a series of more peripheral dependents modifying the former. The core/peripheral distinction of arguments is not always clear (Devine and Stephens, 2006, p. 315). These peripheral dependents include peripheral arguments, so-called direct modification adjectives, relative clauses and relative clause-like adjectives (Cinque, 2010, p. 23-25). The head noun may have internal structures formed with processes resembling phrasal syntax (Di Sciullo, 2005; Scher and Nobrega, 2014). Thus we have the following English example of a nucleus noun phrase: *[the company's]*$_{\textit{subject-genitive}}$ *[three]*$_{\textit{num}}$ *[EXTREMELY AUWFUL]*$_{\textit{prenominal reduced RC}}$ *["attractive*$_{\textit{evaluative}}$ *young*$_{\textit{age}}$*]*$_{\textit{direct modification}}$ *[movie]*$_{\textit{argument}}$ *[superstar"]*$_{\textit{head: compound}}$ *[aboard]*$_{\textit{postnominal reduced RC}}$.

This hierarchical model is not handy for Latin because of the highly variant surface order. Thus, the hierarchical structure of the noun phrase is often ignored in traditional grammar. However, similar to what is done in § 3.2, we are able to show here that the Latin noun phrase has internal configurational and hierarchical structures too.

[10] This is known as a *nominal* in Huddleston and Pullum (2002); the term however is usually used to refer to morphologically nominal parts of speech in the Latin grammatical tradition.

First, the linear order of dependents in a noun phrase appears conditioned by information structure, as shown in the pre- or post-nominal position of arguments (Devine and Stephens 2006, pp. 326-327; Allen and Greenough 1903, p. 396). The head noun is frequently weakly topicalized (scrambling; cf. § 3.2.2.1). This is an information structure operation, because after the head noun has already appeared in the utterance, it is often no longer fronted (Devine and Stephens, 2006, p. 484).

Although these phenomena do not require a configurational analysis, they show that the word order permutation in Latin is not random and at least is compatible with the multiple topicalization-focalization analysis.

Second, the linear order of arguments of deverbalized nouns (where formal structural parallelism between noun phrases and clauses is often easy to spot), resembles that of the corresponding clause (Devine and Stephens, 2006, p. 316-317) and suggests discourse configurationality.

Third, the order between determiners, adjectives, and the head noun and its arguments often reflects their scope, though also strongly influenced by information structure (Pinkster 2015, § 11.75; Devine and Stephens 2006, p. 484). How these factors are reflected in the exact constituent order varies from author to author and from time to time (Devine and Stephens, 2006, p. 484).

We note that even in English, constituent orders do not transparently reflect the relative scopes of noun phrase dependents.[11]

Fourth, among adjectives that seemingly are direct modifications, ones corresponding to the material, etc., tend to appear to the right of ones that require information from the conversational context and are not themselves well-defined properties (Devine and Stephens, 2006, pp. 481-482). This is consistent with the cross-linguistic correlation between the linear adjective order and their scope (Cinque, 2010).

Fifth, there appears to be a tendency for modifiers with intensional meanings to appear before the head noun, while adjectives interpretable without the need to refer to the head noun have both NA and AN orders. Nouns with light meanings are fronted more frequently. These observations are consistent with the following configurational analysis: a fronted (likely weakly topicalized, i.e. "scrambled") constituent is supposed to be semantically less dependent to the rest of the noun phrase it leaves behind, which may hinder the interpretation of intentional modifiers (Devine and Stephens, 2006, p. 485-486).

Latin is a classical language, and detailed tests about the hierarchical structures are not always available and are likely subject to individual variances. However, it should be relatively uncontroversial to state that Latin has discourse configurationality.

3.3.3 Determiners

Latin lacks articles but has rich determiners, traditionally described under the name of *correlatives*. This includes demonstratives, identity pronouns (*ipse* etc.), interrogatives, alternative pronouns (*alius*).

Latin correlatives have complicated inflectional morphology and hence morphologically resemble adjectives. That they are grammatically different from ordinary adjectives is shown by the fact that they tend to precede all other adjectives (§ 3.3.2).

[11]Participle modifiers, for example, seem to have higher scopes, but they regularly appear after the head noun, without reflecting their syntactic positions in the pre-head linear order (Cinque 2010, chap. 5).

Some languages, like English, have a genitive construction in which a genitive phrase is raised out of the nucleus noun phrase and acts just like the subject in a clause (*[his]*$_{\textit{subject-genitive}}$ *play of the national anthem*). Latin doesn't have this kind of constructions.

3.3.4 Numerals

3.3.5 Attributives

In the modifier series surrounding the core argument structure, modifiers that are "direct modifications" and structurally close to the head noun modify it as a whole, just like clausal peripheral arguments and lexical aspect do, and are often semantically non-restrictive[12] and more about eternal properties of the things referred to. Direct modifications usually form a hierarchy in terms of scopes and frequently constituent orders (Huddleston and Pullum 2002, p. 453, [10]; Cinque 2010, pp. 25-34), and are easier to form idioms with nouns they modify (Cinque, 2010, p. 108).

Structurally higher modifiers with wider scope, on the other hand, restrict the reference of the noun phrase, just as structurally higher TAM categories do (Cinque, 2010, pp. 23,25), and they generally look like reduced relative clauses, and are therefore predictive (Cinque 2010, chap. 5)[13] are usually semantically intersective.[14] We note that the innermost direct modification constituents are also intersective in some sense (Devine and Stephens, 2006, p. 476).

3.3.6 Argument structure

Core/peripheral distinction in the argument structure of a noun phrase can be determined by the three parameters listed in § 3.2.5.1, and again the three parameters, despite being occasionally independent to each other, are often correlated, giving us an impression of a unified parameter of core/peripheral distinction (Huddleston and Pullum, 2002, pp. 439-443). The internal makeup of these arguments may vary cross-linguistically: in English they can be noun phrases, nominals,[15] or even adjectives considering (Huddleston and Pullum, 2002, pp. 439-440): thus we have *a [London]*$_{\textit{location direction modification}}$ *[legal]*$_{\textit{argument}}$ *expert*. Just as the manner adverb is closer to the main verb than TAM adverbs, so are these arguments of the head noun

[12] Semantically, some interpretations are also restrictive, as *the beautiful dancers were admitted into the academy* can be understood as 'the aforementioned dancers who were beautiful in their dancing were admitted'. This restrictive meaning however can be derived from pragmatics: the listener is supposed to recall a group of people with the property of being very well-qualified dancers from the conservational context. If the conversational context contains a group of dancers who were not all good at dancing, then *the beautiful dancers* gains a restrictive meaning, but this is not dictated by the syntactic structure. On the other hand, the interpretation of *the dancers that were beautiful* has to be restrictive ('recall the aforementioned dancers – now I'm talking about some of them who were beautiful').

[13] Note that there may be several types of *predicative* dependents, for example in English postnominal and prenominal predictive constituents have subtle meaning differences.

[14] For example, *beautiful* in *a beautiful dancer* may refer to the manner of the dancer dances, or merely be a property of the specific dancer being referred to (in the latter case, its meaning is close to the noun *beauty*). The latter reading is intersective in the sense that the reference of the noun phrase is given by the intersection of the property of being a dancer and the property of being beautiful. The second property has nothing to do with the first, and is equivalent to what a relative clause would contribute to the meaning of the noun phrase.

[15] In the sense in Huddleston and Pullum (2002), not in the Latinist tradition.

closer to the head noun than the direct modification constituents (Huddleston and Pullum, 2002, p. 452, [2]).[16]

3.3.7 Derivation

Finally, at the center of the noun phrase lies the noun stem, which may be a result of derivational morphology. Compounding in nominal derivation should be distinguished from phrasal nominal modification/complementation by tests like coordination and subcategorization – a nominal compound should be more restricted in coordination (*back- and tooth-ache* for example is marginal in English) and may have a subcategorization frame that's totally not predictable from its branches. In Classical Latin real compounding is no longer productive.

3.4 Adjectives and adverbs

Verb-like and

3.5 Lexicon

3.5.1 Parts of speech

Multiple criteria exist for defining parts of speech. Although the criteria themselves look universal, parts of speech divisions we arrive at by applying these criteria often have cross-linguistic variances.

First, parts of speech can be defined according to the distribution of a form in syntactic positions (the head of a noun phrase, the predicator of a clause, etc.). Informally these positions are often themselves named using part of speech terms: hence in describing nominalization we say something "is used as a noun", and the like.

This criterion works well in Latin, as in Latin, it is in general not possible to, for example, put an arbitrary root commonly heading a noun phrase into the predicator position without evidence of verbalization. The situation on the other hand can be much more complicated for Sinitic and Semitic languages. However, based purely on this criterion, it is not immediately easy to identify the adjective class in Latin because Latin adjectives can freely head noun phrases while nouns can be appositive modifiers (TODO: ref on NP structure).

Based on the same criterion, we can distinguish between prepositions and adverbs (but often a preposition has an intransitive counterpart). Conjunctions may be seen as "prepositions for clauses". (Also see § 3.5.3.)

Second, parts of speech can be defined via (realizational) morphology. We easily recognize two adjective classes in Japanese based on this criterion, although their syntactic distributions are largely similar.

In Latin we have two sets of inflectional templates: nominal and verbal morphology. There are also indeclinable forms collectively known as PARTICLES.

[16] It seems Huddleston and Pullum (2002) doesn't consider the possibility of peripheral adjective arguments, which are still to the right of direct modification adjectives.

Nominal inflection is known as DECLENSION and words that decline are NOMINALS, which include NOUNS and ADJECTIVES. Although there is no distinct adjective inflectional template in Latin, that certain words' inflection ending depends on gender while others do not should be enough to recognize an adjective class.

Verbal inflection is known as CONJUGATION. Words that conjugate form a uniform class rightfully called VERBS.

Latin "PARTICLES" include PREPOSITIONS, ADVERBS, INTERJECTIONS, and CONJUNCTIONS, although in modern usage the term *particle* usually refer to grammatical items, and therefore at least some adverbs should be excluded from that class. Some adverbs actually have inflection with respect to *grade* (comparative and superlative).

Evidence from derivation morphology falls in both criteria above.

Third, parts of speech may be defined by form classes in the lexicon. Lexical entries are *lexicalized* in that their syntactic and morphological properties are specified and cannot always be regularly inferred. If a groups of forms share a bundle of grammatical properties, then they are *lexically categorized* into a single part of speech.

We can verify that in Latin, forms with nominal endings regularly appear as heads of noun phrases, thus there is a single, uncontroversial noun class. This class has subclasses, decided by parameters including inflection class (the five Latin declensions), valency, compatibility with single or plural number.

Similarly, a single verb class can be defined by consistency between conjugation endings, agreement patterns, regular appearance as heads of clauses, etc. Sub-classes of the Latin verb class are decided by parameters including the conjugation class (the four Latin conjugations), valency (chap. 9), the event structure (TODO: Compatability with TAM).

We can further recognize an adjective class.

The category ADVERB is often posited as a catch-all category for any lexical words that are not nouns, verbs or adjectives. A majority of adverbs historically originating from fossilized case forms.

It is theoretically imaginable that a form has lexicalized grammatical behaviors that are peculiar and do not fall into any prototypical form class (although more often than not, an approximate part of speech can still be recognized). This phenomenon is attested in Latin, in which a series of irregular nouns and adjectives do not decline or decline in an unusual way.

Some languages have form classes that deviate strongly from the usual parts of speech defined by the first criterion (e.g. Siddiqi et al. 2020, p. 372). In such cases the term *parts of speech* becomes problematic. This is not the case in Latin, and we use the term *form class/category* and *part of speech* interchangeably.

Fourth, semantics is sometimes used to define parts of speech. This is a good start-up strategy but is not a reliable criterion in systematic grammatical description, and suffers from internal theoretical issues. For one, both nouns and adjectives can be seen as describing static properties of objects. We therefore refrain from relying on semantics when studying parts of speech.

3.5.2 Status of composite forms

Parts of speech definition concerns *grammatical* lexicalization, applicable to both non-analyzable forms and composite forms. Cross-linguistically, it has been attested that derived forms may form distinct parts of speech. In Mandarin Chinese, for example, reduplicated adjectives seem to form a distinct form class (Paul, 2014, § 5.3). This is not the case in Latin and derived stems, if lexicalized, can usually be safely classified as nouns, verbs, etc., without marked difference in grammatical properties with those of based words.

Fixing of the *meaning* of a form, i.e. idiomization, is not per se decisive evidence for grammatical lexicalization. Idioms (e.g. *kick the bucket* in English) still obviously need to be stored in the lexicon but they neither need to belong to specific lexicalized form classes nor necessarily have any specific grammatical properties not regularly inferrable from their internal structures.

Further, forms with lexicalized grammatical properties are typically either words or parts of words (§ 3.1.1, § 3.5.4), while idiomization happens to all forms.

Whether lexemes regularly derived should be considered lexicalized entries or just idioms (with fixed meanings but not non-inferrable syntactic behaviors) is a dilemma almost universally faced in lexicography. Diachronically, idiomization often is the first step of syntactic lexicalization and sometimes fossilization (in which the internal structure of the syntactic unit collapses into something different). Both can be attested in Latin, especially in Latin preverbs.

3.5.3 Function words, closed categories

Syntactic fossilization may eventually end with grammaticalization. A prototypical example is that complement-taking verbs become auxiliary verbs, which become a part of the clausal grammar and have to appear in a given order.

In theory, grammatical forms (affixes or auxiliaries or particles) should not be seen as a part of the lexicon (and hence do not even necessarily need part of speech tags). In practice this is less clear-cut.

First, whether grammaticalization has occurred is not always uncontroversial (and the process can also be asynchronous). Danckaert (2017, p. 140) for one argues that *possum* and *debeo*, commonly taken to be full verbs, had probably started to become grammaticalized in Classical period. Due to the lack of acceptability tests and psycholinguistic data (in e.g. acquisition or agrammatism), the content/function distinction is especially hard for Latinists.

Second, certain languages have a rather massive inventory of grammatical forms. and for descriptive conveniences, part of speech classification of them as if they were content words turns out beneficial. This is the case in Latin, which has an abundance of correlatives and particles (regarding which the question of grammaticalization also lingers).

"Parts of speech" of grammatical forms may be roughly divided into those that are in roughly the same positions of content words (e.g. personal pronouns, correlatives), and those that are purely grammatical markers. This distinction is however not always uncontroversial as it depends on the details of linguistic analysis.[17]

TODO Latin has pronouns and pro-adverbs; depending whether we treat prepositions as heads: though prepositions are often said to be markers of a periphrastic case system, the semantics carried by certain Latin prepositions are too complicated for a case system, and these prepositions can be understood as adverbs selecting a complement. TODO: adverb modifying PP This is also the case of adverbs: some adverbs

[17] One may for instance ask in which way an auxiliary verb is in a verbal position. Even further, one can argue that noun endings and verb endings "occupy positions content words occupy".

seem to be periphrastic markers of TAM categories and therefore may be considered as a part of the grammar, while others seem to carry "real" meanings

In some languages, like Japanese, there exist form classes that are closed but are clearly content word classes (which often occasionally do admit new members, albeit slowly). Latin does not have such word classes.

3.5.4 Half-words

In some languages, there are "half-words", known also as combining forms or bound roots, that appear in phrasal syntax-like environments (Di Sciullo, 2005; Scher and Nobrega, 2014) and are clearly lexicalized in that they have preferential positions in compounding too that cannot be regularly inferred, but are not lexemes. Examples include English half-nouns *geo-* or *-logy*.

Latin is the origin of many combining forms in modern European languages. It has far less combining forms in itself, but does

3.5.5 Lexicography

We have seen that lexicography of a language should be based on language-specific analyses. Part of speech tags should consider both syntactic and morphological properties and should faithfully represent correlation between these properties. Although grammatical markers in theory are not a part of the lexicon, they are often included into dictionaries for convenience and for blurry lexical/functional distinctions.

Fig. 3.1 is a visualization of the classification of Latin word classes.

How a dictionary should be organized also depends on the peculiarities of a language. Dictionaries of Semitic languages and Sinitic languages often revolve around roots: the former have regular derivations that can be analyzed by (uncategoriezed) roots combined with vowel patterns, while the latter have regular compounding, so root-based lexicography is beneficial. In other languages, Latin and English included, a word-based (or perhaps we should say lexeme-based) dictionary is preferred because derived forms often have obscured meanings if not structures (for example, the meaning of the prefix *re-* is not always clear), and uncontroversially productive uncategoriezed roots are rare. We note that this likely was not true in Proto-Indo-European.

3.6 Reading Latin

The morphological richness (and the scrambled constituent order) makes Latin hard to read especially for people whose first languages are, say, English or Mandarin. Whenever unsure about a sentence, do the follows:

1. Skim over the words and label the stems that can be easily recognized.

2. Skim over and circle uncontroversial grammatical items, like inflectional endings and prepositions. It's OK to be unable to interpret them immediately (and we need the steps below).

3. According to these grammatical items, segment the sentence into large parts: noun phrases, relative or complement or adverbial clauses, verbal system, etc.

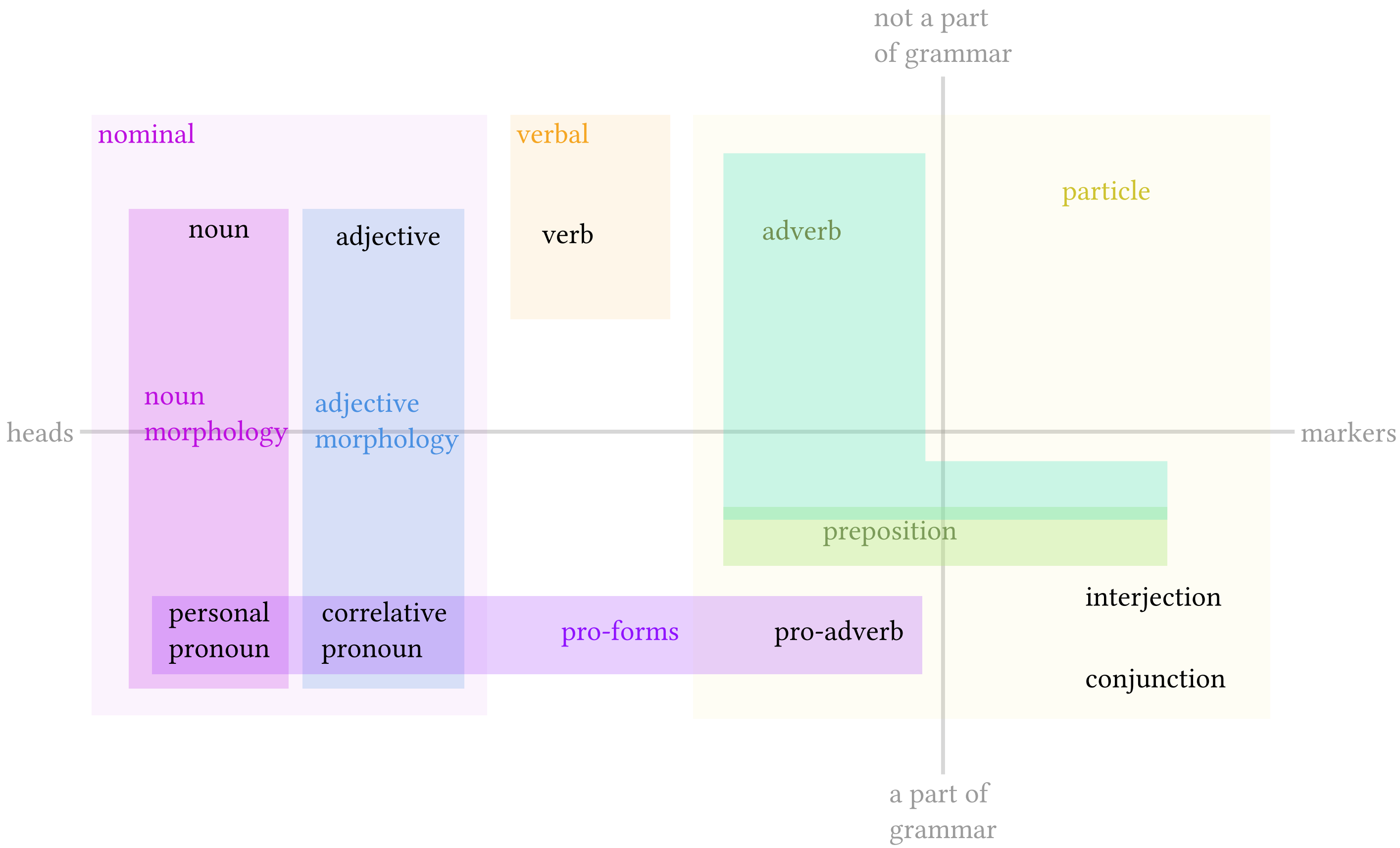

Figure 3.1: Latin word classes

4. Choose a grammatical item and tentatively give a list of possible features it carries. For example, seeing *-v-* in a verb usually means it's based on the perfect stem (§ 8.2.3); *-um* may be second declension accusative, but there are other possibilities (Table 5.2).

5. Use constraints like "the preposition *in* licenses the accusative case or the ablative case" to narrow the possibilities identified above.

6. Draw unfinished dependency arrows: for a verb, draw arrows pointing to the subject and/or the object; for a nominative adjective, draw an arrow pointing to the modified head noun. But note that it's possible that the subject is dropped, or there is no head noun (compare English *the poor*). Then try to pair the arrows.

Repeat the above procedure and finally the sentence can be understood. This procedure is demonstrated in § 15.3.1.

Chapter 4

Pro-forms

Pro-forms can be divided into pronouns and correlatives.

4.1 Personal pronouns

Latin pronouns are complete noun phrase themselves: no attributives should be attached to them. Pronouns are declined for case, gender and number, and they also can be governed by prepositions.

It should be noted that what counts as third person pronouns is not uncontroversial among reference books regarding Latin; some place the reflexive *sui* series into the third person column in the table of personal pronouns, citing the fact that the *is, ea, id* series mark the gender category, while the first and second person pronouns do not.

4.2 Reflexive pronouns

It can be seen that *se* lacks nominative forms. In some circumstances, the reflexive *se* looks like the subject; but these cases are more appropriately analyzed as reflexive usages of transitive verbs (1).

(1) ...necessarium est primo investigare de ipsa sacra doctrina, qualis sit, et ad

quae [se extendat]

'It is necessary to first investigate into this sacred doctrine, of what kind it would be, and what it would extend itself to (i.e. its nature and extent).' (*Summa*, I q. 1 pr.)

4.3 The "basic" correlatives

Latin correlatives can be classified in Table 4.1; the rows correspond to their immediate roles in the clause ("basic" i.e. noun phrase heads, place, source, time, etc. that are otherwise expressed by prepositional phrases), while the columns correspond to their meaning and/or relation with a more precedent in a higher position, like whether the word is a demonstrative or maybe a relative pronoun.

The form of correlatives in the "basic" can be summarized as the follows:

Table 4.1: Classification of Latin correlatives

		"question"			demonstrative						
		interrogative	relative	indefinite relative	proximal	medial	distal	identity	indefinite	collective	negative
nominal head	basic										
	dual										

- *Indefinite.* They may be homonyms of *qui* or *quis*, or they may be formed by adding *ali-* or *-dam* to *qui* or *quis* (with possible internal alternations), or maybe they are formed in the format of *ali-quis-X-quam*.
- *Indefinite relative.* Formed by reduplication of *quis* forms.
- *Other.* Always take a form similar to *alter* or *alius*.

Here is a guide for quickly determining the meaning of a "basic" pro-form:

- What starts with *ill-* is a distal demonstrative.
- What starts with *ist-* is a medial demonstrative.
- What starts with *h-* is a proximal demonstrative.
- What starts with *s-* is a reflexive pronoun, or maybe a reflexive possessive determiner like *suus*.
- What starts with *qu-* is an interrogative pronoun or a relative pronoun or an indefinite pronoun or an indefinite relative.
- What starts with *null-* is a negative pronoun.
- What starts with *omn-* is a collective pronoun.
- What starts with *ips-* is an identity pronoun ('this exactly ...').
- What starts with *alter-* is an "other" pronoun.
- What starts with *ali-* may be an "other" pronoun, or an indefinite pronoun.
- What *ends* with *-dem* and starts with a third person pronoun is an identity pronoun; sound changes are possible. For example *īdem* or *eandem*.
- What *ends* with *-dam* and starts with *q-* is an indefinite pronoun.

Some forms can be easily confused with correlatives. They include:

- *quod* used to introduce a finite complement clause.
- *quia* as a conjunction of a reason clause.

4.3.1 Demonstratives

Latin has proximate, medial, and distal demonstratives: the first refers to something close to the speaker, the last refers to something far away, and the second refers to something in between, probably near the listener.

4.3.2 Interrogative and relative pronouns

4.4 Degree

The *tam quam* pair may be seen as a pro-form that takes the place of the degree expression of a clause.

Chapter 5

Nominal morphology

5.1 Word structure

Latin nouns decline for case and number, and also the inherent feature of gender. These features spread to other morphologically nominal words in the noun phrase by agreement.

The structure of the Latin noun is the stem plus inflectional ending with the case and number categories fused into one suffix (Table 5.1).

Table 5.1: Latin noun word structure

noun form	stem	ending
rosam 'rose-ACC.SG'	*rosa-*	*-m*
hominis 'human-GEN.SG'	*homin-*	*-is*

5.1.1 Stem alternations

A phenomenon shard by many IE languages is the presence of a thematic vowel at the end of the stem: this, in Latin, is e.g. the stem-final *a* sound in *rosam*. The thematic vowel indicates the declension class of a noun lexeme (Oniga, 2014, p. 45).

The thematic vowel undergoes changes: it may be deleted and its length may shift.

Stem alternation thus leads to a discrepancy in the literature in segmenting a noun into its stem and its ending.

One convention is to find the common part of all case forms of one nominal lexeme, and thus *rosam* 'rose-ACC.SG' is analyzed as *ros-am*. Allen and Greenough (1903, p. 17) documents the paradigms of all the five declension classes in this approach but cautions that the "stem" defined in this way is not necessarily the stem and is, in their terminology, a *base*.[1]

Modern analyses treat the thematic vowel a part of the stem (Oniga, 2014, pp. 45, 63). Thus, the first conjugation singular accusative ending *-am* is *-a-m*, with *-a-* being the thematic vowel, and we attach the thematic vowel back to the stem in the narrow sense *ros-* and redefine the complex *ros-a-* as the stem of *rosam*.

[1]Needless to say this term has a rather different meaning in derivational morphology in some works, like Huddleston and Pullum (2002).

This catches obvious parallels between different declensions (e.g. first declension *-am* vs. second declension *-um* vs. third declension *-em*). Further, we notice that the analysis that treats the stem as the invariant part of the word fails to identify the stem of *hominis*, in which internal alternation appears.

5.2 Noun inflection classes and endings

There are five inflection classes for Latin nouns, known as the five declensions. Table 5.2 enumerates attested endings and their declensions. For convenience, the thematic vowel is included as a part of the ending (cf. § 5.1.1).

Table 5.2: Declension endings; Roman numerals are declension classes

ending	declension
-a	I, SG.NOM, SG.VOC; IIN, PL.NOM, PL.ACC, PL.VOC; IIIN, PL.NOM, PL.ACC, PL.VOC
-ā	I, SG.ABS
-ae	I, SG.GEN, SG.DAT, PL.NOM, PL.VOC;
-am	I, SG.ACC
-ārum	I, PL.GEN
-ās	I, PL.ACC
-e	IIM, SG.VOC; IIIFMN, SG.ABS
-ē	V, SG.ABS
-ei/-ēi	V, SG.GEN, SG.DAT
-em	IIIFM, SG.ACC; V, SG.ACC
-ēbus	V, PL.DAT, PL.ABS
-ērum	V, PL.GEN
-ēs	IIIFM, PL.NOM, PL.ACC, PL.VOC; V, SG.NOM, SG.VOC, PL.NOM, PL.ACC, PL.VOC
-ī	IIM, SG.GEN, SG.VOC, PL.NOM, PL.VOC; IIN, SG.GEN; IIIFMN, SG.DAT, SG.ABL
-ibus	IIIFMN, PL.DAT, PL.ABS; IVFMN, PL.DAT, PL.ABS
-is	IIIFMN, SG.GEN
-īs	I, PL.DAT, PL.ABS; IIMN, PL.DAT, PL.ABS
-ō	IIMN, SG.DAT, SG.ABS
-ōs	IIM, PL.ACC
-ōrum	IIMN, PL.GEN
-r	IIM, SG.NOM, SG.VOC
-ū	IVFM, SG.ABS; IVN, SG.NOM, SG.DAT, SG.ACC, SG.ABS, SG.VOC
-ua	IVN, PL.NOM, PL.ACC, PL.VOC
-uī	IVFM, SG.DAT
-um	IIMN, SG.ACC; IIN, SG.NOM, SG.VOC; IIIFMN, PL.GEN; IVFM, SG.ACC
-us	IIM, SG.NOM; IVFM, SG.NOM, SG.VOC
-ūs	IVFM, SG.GEN, PL.NOM, PL.ACC, PL.VOC; IVN, SG.GEN
-uum	IVFMN, PL.GEN

There are several possible cases where something seeming to be a noun case ending turns out to be something else. To name a view:

- The ending *-as* is not always the first declension plural accusative ending; *-a-* could be a part of the stem of a third declension noun. Consider *vēritās*.

- The ending sequence *-io* can'be be found in Table 5.2 and we may hurry to the conclusion that it's the third declension abstract noun ending *-io* in the nominative or accusative case. Not necessarily – it can also be *-ium* in the dative or ablative case (when the macron symbol for long vowels are not used).
- A word ending with *-us* could be an adverb: *-tus* is an adverb suffix.

5.3 Declension of regular nouns

The highly complicated list of noun endings (Table 5.2) may be compared with the much briefer nominal paradigm in PIE. A comparison between reconstructed PIE noun and pronoun endings and early Latin endings is given in Clackson and Horrocks (2011, p. 14).

Table 5.3: PIE nominal case endings

	singular	plural
NOM.M/F	*-*s*	*-*es*
NOM.N	*-*m*/∅	
ACC	*-*m*	*-*ms*
GEN	*-*es*/ *-*os*	*-*om*
DAT	*-*ei*	*-b^h*os*/ *-*ōis*
ABL	*-*ōd*/∅	

Box 5.1: PIE ending

- What's the neutral plural ending?
- Stem's selection of endings in PIE

5.3.1 The first declension

5.3.2 The second declension

The stem-final thematic vowel in the second declension is historically *-o-*. Because of sound changes, the NOM.SG ending becomes *-us*,

5.3.3 The third declension

The third declension is a big tent containing several subclasses. Nominative singular endings attested in the third declension include *-s*, *-t*, *-x* (i.e. *-cs*) (Allen and Greenough, 1903, § 53).

5.4 Derivational morphology

Inside the stem we find a list of

Chapter 6

Cases and prepositions

The roles of the five cases are not symmetric.

On the other hand, the rest cases are *inherent* cases: they have direct semantic interpretations – "source" or "target" or ...– themselves, and once an inherent case is assigned to an noun phrase, the latter is "sealed" just like a prepositional phrase: the change of the outside syntactic environment doesn't change anything inside. Sometimes inherent cases are realized as prepositions.

Historical evolution of case morphology From PIE to Romance languages, the case system in Latin has progressively eroded. There are two ways to simplify a case system: it's possible that two morphological cases are merged altogether, which means the relevant grammatical relations are no longer realized in nominal morphology (e.g. § 6.0.2); it's also possible that the structure of the case paradigm is retained, although for some declension classes, some of the case forms are regularized (e.g. § 6.0.1).

6.0.1 The nominative

Historical evolution in Vulgar Latin The first declension plural nominative ending *-ae* started to be replaced by the accusative *-as* starting at no later than first century A.D. but arrived at Gaul only after the fifth century. Note that this replacement is not equivalent to elimination of morphological marking of the subject/object distinction: the *-us*/*-um* alternation is still kept (Herman, 2010, p. 55). The motivation may be analogy of the neutralization of the nominative and accusative endings in the third declension, and/or analogy of Oscan and Umbrian nominative plural ending *-as*.

6.0.2 The accusative

In Vulgar Latin A general tendency in Vulgar Latin is the ablative in prepositional constructions being replaced by the accusative (Herman, 2010, p. 53). Similarly, internal arguments which should be genitive or dative or ablative tend to be accusative in Vulgar Latin (Herman, 2010, p. 54). This suggests the accusative being recognized as the "default case".

6.0.3 The dative

The dative case is an inherent case assigned to the benefactive, the experiencer, and the purpose (Oniga, 2014, p. 251); thus it's the case assigned to the indirect object (§ 9.6.1) as well as many adjuncts TODO

6.0.4 Ablative

so its distribution is similar to the dative case: it appears in source and instrument indirect objects (TODO) as well as various adjuncts.

A source argument can be the position from which something moves (**ablative of source**) or the source in a separation event (**ablative of separation**: 'remove', 'deprive'), or the place where something comes into being (**ablative of material**, 'birth', 'origin'), or the cause of something ("the source of the event", **ablative of cause**); the agent in the passive voice possibly comes from one of the figurative use of the ablative as well.

Chapter 7

Adjectives and adverbs

The adjective class and the adverb class are linked together by several factors: the adjective phrase and the adverb phrase are both prototypical modifiers, often with parallel structures; they both have the category of degree; adverbs can be formed regularly from adjectives.

7.1 Declension of regular adjectives

Peripheral arguments may also be regarded as adverbials. This chapter, however, is mainly about mean, TODO

7.2 Arguments of adjectives

7.2.1 Dative

(1) … consubstantialem Patri …
consubstantial-SG.ACC Father-SG.DAT
'consubstantial with Father' (Nicene Creed)

7.3 Comparative construction

7.4 Prepositions

7.4.1 Are prepositions function words or content words?

The preposition class usually shows great diversity both within a language and among different languages, and what are known as prepositions may be lexical words or function words (e.g. Garzonio et al. 2021). In certain languages, some prepositions are obligatorily selected in certain grammatical constructions and have to form a hierarchy when two or more of them appear together, and hence may be seen as a case particle: they are phonological realizations of grammatical relations, not lexical words. Other prepositions are just adverbs with complements. Several case particle-like prepositions have been discussed above;

> **Box 7.1: Latin preposition and adverbs**
>
> Can *ex* or *in* be analyzed as adverbs? If so they should be able to be modified by other adverbs?

English *of* is used to introduce the object in nominalization and *by* is used to introduce the deep A argument in passivization. We can therefore say that the first is an analytic genitive case marker, and the second is an analytic instrument case marker.

In Latin, one use of *ab* falls in this category: personal agents in passive clauses are always preceded by *ab*. The ablative case governed by *ab*, in this use, marks nothing: this resembles the fact that English perfect auxiliary *has* selects a past participle or the Japanese verbal conjugation suffixes add CONTINUATIVE or GERUND endings to the verbal complex before them, which can be explained by diachronic reasons.

Preposition sequence in directional constructions In directional constructions, stacking of case markers and/or prepositions in a fixed order and with a finite number of possibilities indicate that the prepositions have already been grammaticalized. This is the case in Finnish or, to a lesser degree, in English (Svenonius 2010; consider English *from behind the wall*).

In Classical Latin there is no clear example of this type: no preposition stacking is allowed in Latin, so coexistence patterns cannot be used to establish a lexical/functional distinction. We however note that directional prepositions in Latin seem to always encode the path and never the properties of the location. Semantically, *ex* encodes a path going away from a given region, while *in* encodes a path going into a given region, and the status of the location as the goal or the source is marked by its case. Tentatively we can analyze *ex* and *in* as grammaticalized markers of the type of the path in the directional construction.

In early Christian Latin, the form *abante faciem domini* 'from in front of the face of Lord' is attested (Herman, 2010, p. 24), and the path-figure-ground hierarchy is perfectly demonstrated here. This shows that at least in some sub-elite varieties of Latin, *ab* and *ante* are grammaticalized directional markers.

Preposition as modifier in another prepositional phrase It is also possible that a preposition is an adverbial modifier of an existing prepositional construction (Botwinik-Rotem and Terzi 2008; c.f.English *the boat drifted [**down** from **up** above the dam]*). This case seems absent in Latin.

Comitative The preposition *cum* regularly marks the comitative construction, both in verb frames and in comitative noun phrases (NPs).

7.4.2 Prepositions as transitive adverbs

Some Latin prepositions are likely as lexical as other lexical words. They are "transitive adverbs".

The form *circum* appears in several contexts and means a (static) location in space or a dynamic path depending on whether a path is implied by the verb; in all these cases, however, the meaning 'around sth.' is always kept. This is very unusual if it is understood as a purely grammatical orientation particle: homophonous grammatical particles appearing in two contexts in principle can have rather different meanings. It is likely a root surrounded by noun phrases expressing the GROUND and the FIGURE in a spatial relation, resulting in a locational expression Mare (2018).

7.4.3 Intransitive prepositions

Many prepositions can also be used without any complements and may thus classified as an adverb; they can also appear as preverbs.

The preposition/adverb distinction

Chapter 8

Verb morphology

8.1 The structure of the inflectional paradigm

8.1.1 The verb template of the finite paradigm

For a heavily inflected language like Latin it's an appealing idea to start the discussion on clause structure with a surface-orientated analysis of verb structure. The morphological template is shown in Fig. 8.1; some inflected Latin verbs and their parts are shown in Table 8.1. Traditionally, the verb is divided into the **stem** and the **ending**. Derivation in Latin is predominantly preverbal, and hence the conjugation is mostly about the final lexical morpheme in the verb stem, which is represented as the root in Fig. 8.1. There may be a perfect suffix after the root. Components of the verb ending include the **tense and mood suffix** (also known as the **tense suffix**), and the person, number and voice marker, which is called the **personal ending** here, following the terminology in Allen and Greenough (1903, § 165).

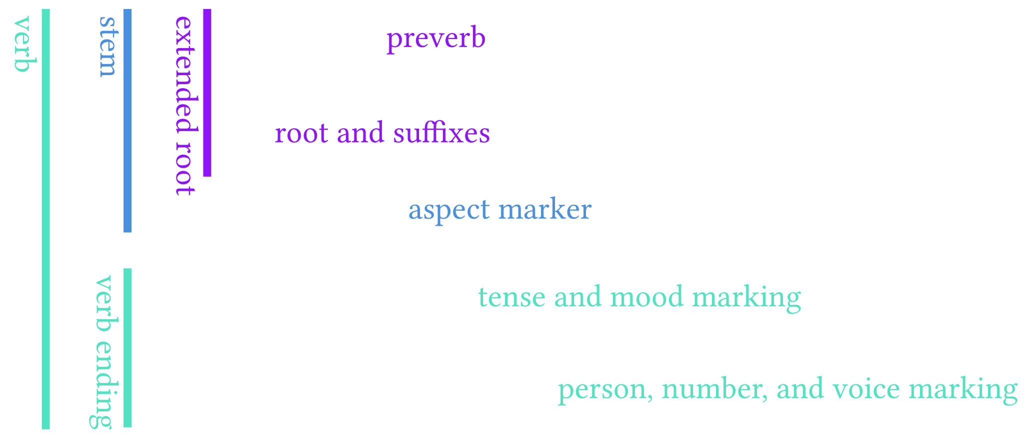

Figure 8.1: The template of Latin verbs. Indentation means linear order and not necessarily constituency structure.

The core stem assumes semi-regular alternations. The perfect marker is often but not always *-v-* (§ 8.2.3). Although the aspect marker is a part of the TAM marking (§ 8.1.2), it somehow is much more closely attached to the root in the morphophonological realization of the verbal system: how it is realized is not completely predictable, and therefore the two finite stem forms – the present stem and the perfect stem – are both required for complete characterization of the paradigm of a regular verb. To fully decide the paradigm we still need a further stem, the supine stem (§ 8.2.1).

Contextual allomorphs also exist for other morphemes in Fig. 8.1 (Embick and Halle, 2005, p. 11), as can be observed in Table 8.1. The most salient change is that an alternating vowel may appear between the root and the aspect suffix (if any), which is also known as the **thematic vowel**. It is the residue of the proto-Indo-European (PIE) ablaut and labels the conjugation class (§ 8.2.4), and is subject to morphophonological contextual alternation (§ 8.3.1). The tense affix is determined by "tense" in the sense of traditional Latin grammar, i.e. both tense and aspect (Table 8.3), and has dependence on the conjugation class (TODO: ref). The perfect tense also has its own personal endings (Table 8.4).

> **Box 8.1: More than one thematic vowel?**
>
> Phenomena similar to the thematic vowel – vowel alternation with no morphosyntactic significance between two morphemes – can be observed between the tense suffix and the personal ending (as in *amā**bu**ntur*; § 8.3.3) as well. We may go further and say the -*imus* ending seen in first person active indicative perfect, as opposed to -*mus* observed in other cases, contains a thematic vowel *i* after the perfect marker -*v*-. Some generalize the concept of thematic vowel and stipulate a thematic vowel position after *every* morpheme Embick and Halle (2003). This is also motivated by evidences from some descendants of Latin (Oltra Massuet, 1999; Oltra-Massuet and Arregi, 2005). Whether this is needed for describing Latin however is

Table 8.1: Examples of Latin finite verbs

verb form	stem			tense and mood	pers. ending
	extended root		aspect marker		
	ext. rt.	thematic v.			
amō	*am*				*ō*
laudāmus	*laud*	*ā*			*mus*
olēvimus	*ol*	*ē*	*v*		*imus*
amāveris	*am*	*ā*	*v*	*eri*	*s*

Below I discuss the subsystems in Fig. 8.1.

8.1.2 TAM categories

Two subsystems can be observed in the tense and mood suffix in Fig. 8.1, which are of corse, tense and mood. Six tense morphological categories can be recognized in total: present, imperfect, perfect, pluperfect, future, future perfect. There are two mood morphological categories: indicative and subjunctive; sometimes the imperative is recognized as the third mood. These categories can be established based on sole morphological basis.

We need to note that the morphological marking strategies of TAM do not transparently reflect the underlying abstract TAM features (as is illustrated in Table 8.2). Following the example in Grimm (2021), in this note, I use small capitals for the names of attested surface *morphological realizations* of TAM, and I use default fonts for abstract TAM values. Some other grammars, like Jacques (2021); Friesen (2017), use initial capitals for the former.

Furthermore, two constructions with *syntactically* different abstract TAM features may have comparable *semantics*. Examples include that if readers of a text know it's about historic events, using present or past tenses in the text will not influence its meaning, and that the distinction between simple past and present perfect isn't always considerable. Sometimes this leads to morphological merger of TAM values (Table 8.2). Choosing one of several constructions with almost identical meaning sometimes follow stylist conventions (§ 8.4).

Marking of primary and secondary tenses Abstract TAM categories marked by TENSE morphology includes at least primary tense (past, present, future) and secondary tense (anterior i.e. perfect v.s. simultaneous i.e. plain): the PLUPERFECT, the PERFECT, and the FUTURE PERFECT all have a perfect meaning.

Point of view aspect and its combinatorics with secondary tense It seems the PERFECT can also be used to refer to any incident in the past, while the IMPERFECT can also describe something happening in the past. The distinction between the two is likely that point of view aspect (imperfective i.e. progressive v.s. perfective): the IMPERFECT has a progressive meaning.

After enumerating possible combinations of the imperfective-perfective distinction and the secondary tense, we find that among four possible options, three are directly morphologically marked: plain (simultaneous, perfective), imperfective (simultaneous, imperfective), perfective (anterior, perfective). The three combinations are often known as the three *aspects* in Latin linguistics. In the past paradigm they correspond to the PERFECT, the IMPERFECT, and the PLUPERFECT, respectively.

The fourth combination, the combination of the imperfective and the anterior aspectual values, i.e. "progressive perfect" (*have/had been doing* in English), does not have an independent morphological marker in Latin. This may originate from the fact that in most situations, the perfect secondary tense presumably has a perfective meaning, as observing an event from the outside (perfect i.e. anterior secondary tense) usually means the internal makeup of the event does not matter (perfective aspect); indeed, English *have/had been doing* is mostly used to convey the meaning that the situation being described is still going on at the reference time, although this is an implication and not a proper part of the meaning of the perfect imperfective (e.g. in *we've being working on this project for at least two years; we finished it today, and today, we proudly present it to you.*, the project development was already finished at the speech time).

The TAM values of morphological TENSES The composition of present, past and future tenses and plain, imperfect and perfect "aspects" gives nine options; these options are further reduced to six because of several additional mergers.

First, the past simple and the present perfect are identified with each other, possibly because of obvious semantic closeness. Also, the imperfective and simple "aspects" are fused when the primary tense is present or future, or in other words, the point-of-view imperfective/perfective distinction is not reflected in the present or future paradigm: only the spontaneous/anterior (i.e. simple/perfect) values of the secondary tense is morphologically relevant.

The six remaining TENSE forms and their TAM values are shown in Table 8.2. In the Latin grammatical tradition, the term **tense** usually refers to these six options, instead of the past/present/future system.

Table 8.2: Latin TENSES

	past	present	future
imperfective	IMPERFECT	PRESENT	FUTURE
simple	PERFECT		
perfective	PLUPERFECT	PERFECT	FUTURE PERFECT

The usage of the six TENSE categories have peculiarities that cannot be completely summarized into Table 8.2; these are discussed in § 8.4.

The function of MOOD markers Similar fusion between categories is shown in the category of MOOD.[1] It's the fusion of morphologically marked clause type (declarative and imperative) and morphologically marked modality. The verb morphology of interrogative clauses is exactly the same as declarative clauses: the interrogative clause type is marked by the existence of interrogative *pro*-forms. Thus, there are three moods in finite clauses in Latin: INDICATIVE, SUBJUNCTIVE, and IMPERATIVE. The INDICATIVE is the fusion of the declarative/interrogative clause type and the realis modality. The SUBJUNCTIVE mood is the fusion of the declarative/interrogative clause type and the irrealis modality. The IMPERATIVE is basically the imperative clause type: it doesn't allow modality marking. Sometimes people say the infinitive is the fourth mood, though it's a non-finite clause.

8.1.3 Agreement

Latin is a typical nominative-accusative language, both morphologically and syntactically. In finite clauses, there is subject-verb agreement: the number and person of the subject is marked on the main verb. In the case of periphrastic conjugation, the features are marked on the copula.

8.1.4 Compatability of categories

There is no FUTURE tense and FUTURE PERFECT tense in subjunctive clauses, probably for the semantic reason that the future tense already contains certain sense of modality (an event predicted to happen), and thus is not compatible with the SUBJUNCTIVE mood. The IMPERATIVE mood is not compatible with other TAM markings except the PRESENT tense and the FUTURE tense. It's still compatible with the voice category, and allowed persons are second person singular/plural with the PRESENT tense, and second/third person singular/plural with the FUTURE tense. The absence of first person is also probably from semantic origin.

In conclusion, the categories involved in the finite verb paradigm of Latin are shown in Fig. 8.2. Here mood and tense are realized in one morpheme, and voice,

[1] Dixon (2009) only calls the category of clause type *mood*. Huddleston and Pullum (2002), on the other hand, calls syntactic modality mood and uses the term *modality* for pure semantics. Different linguists use the term *mood* and *modality* in radically different ways. In this note I just focus on the common practice in Latin grammar study. In terms of general linguistics, the Latin *mood* is a mixture of clause type (the real mood) and modality.

person and number are realized in one morpheme. The paradigm is realized synthetically in all circumstances except in passive voice and perfect tense. In that case, the verb conjugation is realized like the English passive, i.e. via a copula and the perfect passive participle.

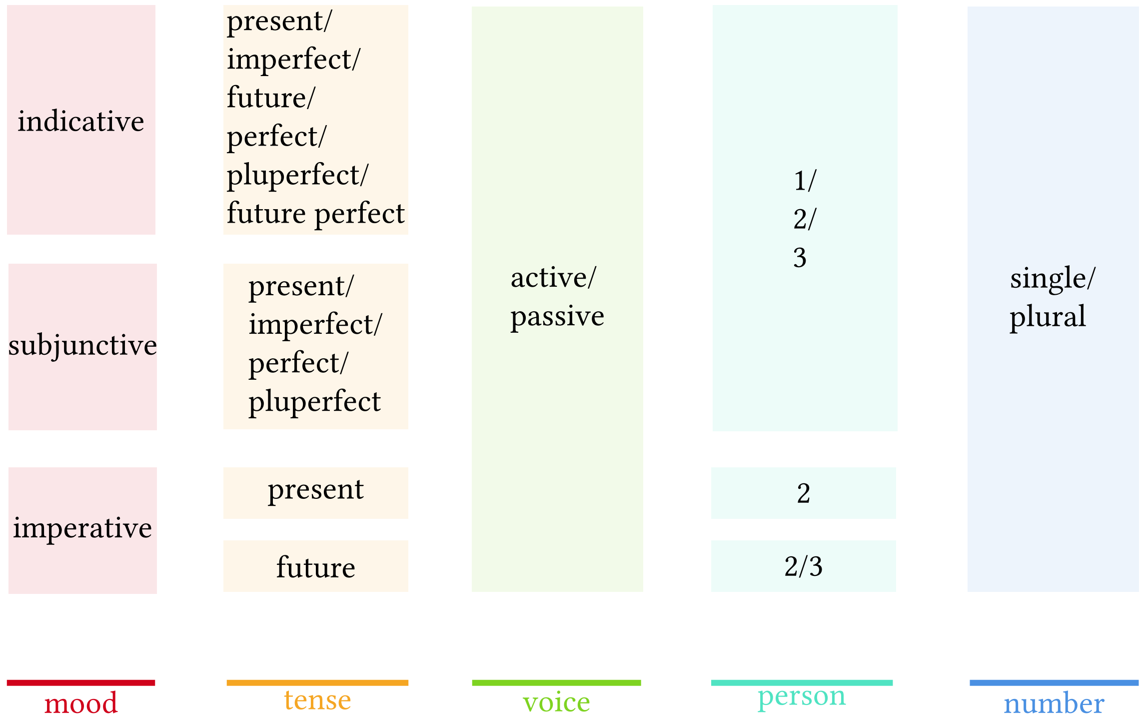

Figure 8.2: Categories in the finite paradigm

8.1.5 The non-finite paradigm

According to the morphology, Latin non-finite verb forms can be classified into the infinitives (§ 8.5.1) and the nominal forms (§ 8.5.2), the latter having noun-like or adjective-like morphology. Non-finite verb forms don't agree with the subjects they take, so there is no number or person category marked on them in the same way as Fig. 8.2, though for nominal verb forms there are number and person categories marked in the same way as the nominal morphology.

The infinitives include present active, present passive, perfect active, perfect passive, future active, and future passive infinitives. The latter three are realized periphrastically (Fig. 8.3).

The nominal verb forms include the **simple active**, the **perfect passive** (often just called the perfect participle), and the **future active** participles, the **gerund**, the **gerundive** which is also known as the **future passive** participle, and two supine forms. The **first supine** is identical in the form to the singular neutral accusative perfect participle, without any reference to the number category of any argument it takes. The **second supine** is identical to the singular neutral ablative or dative past participle, also with no inflection with respect to the number category of any argument it takes. The gerund is morphologically the singular neutral of the gerundive (§ 8.5.2.5).

I keep the traditional notion that the two supines are better recognized as separate morphological inflection forms, instead of two specific usages of the perfect participle and the past participle: although the two supines are identical to two other inflection forms in the surface form, that they don't have number agreement or case inflection means they are products of morphological devices that differ from the participle constructions.[2]

In Classical Latin, the gerund and participle forms are significantly more noun-like than their counterparts in English, and this also justifies the term *nominal form*, because they are not far from prototypically nominalization: although they are still modified by adverbs, they are unable to take arguments. In Ecclesiastical Latin, the so-called nominal forms are more verb-like (TODO: ref), being able to take arguments, and are therefore no longer "nominal".

8.1.6 The two periphrastic conjugations

There are two additional periphrastic constructions beside the periphrastic forms in the perfect, passive part of the paradigm. In the first construction, the copula *sum* with arbitrary person, number and TENSE inflection (*including* the perfect TENSES) is placed together with the future active participle, to express the meaning of 'I/you/he/she/it/they are going to do something', TODO: a inchoative feeling? This construction is always active in voice. In the second construction the future active participle is replaced by the future passive participle; the meaning is 'something should be done'.

The present active participle is almost never used in periphrastic conjugation. It can be seen that in periphrastic conjugation, the roles of future active participle and future passive participle and the role of passive perfect participle are not comparable: the latter dictates the secondary tense, i.e. it's perfect, as well as the voice; the former dictate voice and a "lower" aspect/modality category (inchoative or obligation), but not the secondary tense; and in particular, the term *future* is kind of (although not really) misleading: the two future participles carry aspect and modality values with them, which are semantically related to the concept of future; but the future primary tense is not involved.

8.2 Formation of stems

8.2.1 The three verb stems

Latin shows stem alternation that is not completely predictable. All verb forms can be obtained by three stems (Allen and Greenough, 1903, § 164), if the verb is regular:

- The **present stem**, which, after attached with proper endings, forms
  - The PRESENT, IMPERFECT, and future forms, regardless of whether they are indicative or subjunctive, active or passive. (There is no future or future perfect subjunctive).
  - All the imperatives.
  - The present infinitives, active and passive.

[2] Similarly, the infinitive (or the "plain form", since the infinitive is actually a label of clauses) is recognized as a form independent from the present form in English in Huddleston and Pullum (2002, p. 74).

    - The present participle, the gerundive, and the gerund.

- The **perfect stem**, which, after attached with proper endings, forms
    - The perfect, pluperfect, and future perfect active, indicative or subjunctive. Again, there is no future or future perfect subjunctive. Note that the passives are *not* formed by the perfect stem.
    - The perfect active infinitive. (Or the perfective infinitive active, since infinitive is considered as a mood by some people.)

  Note that the perfect passive participle is *not* obtained from the perfect stem.

- The **supine stem**, which, after attached with proper endings or used together with proper forms of *sum*, forms
    - The perfect passive participle, which, by being used with proper forms of *sum*, forms
        * The perfect, pluperfect, and future perfect passive forms, indicative or subjunctive. Again, there is no future or future perfect subjunctive. This is periphrastic conjugation: it is done by using proper forms of *sum* with the perfect passive participle.
        * The perfect infinitive passive.
    - The future active participle, which, used together with *esse*, makes the future active infinitive.
    - The future passive infinitive, by being used together with *īrī*.

This process is summarized in Fig. 8.3.

In a dictionary, typically the stems are not directly given – which are given are representative verb forms, from which the stems and the conjugation class can be inferred. The reasons are the follows. First, for fluent users, recording actually attested word forms is easier compared with the morpheme-based "anatomized" approach. Second, Latin has four conjugation types, and hence the three stems themselves aren't sufficient to decide how to conjugate the verb: more information is needed, and by storing already conjugated verb forms, the conjugation class can be decided by observing the endings. What are stored are the following **principal forms**, from which the three stems and the conjugation class can be solved out (Allen and Greenough, 1903, § 172):

1. *The first-person present active indicative*: formed from the present stem.

2. *The present infinitive*: formed from the present stem. By observing its ending, the conjugation class can be decided, and by comparing with the first principal form, the present stem is obtained.

3. *The first-person perfect active indicative*: showing the perfect stem.

4. *The neutral accusative past participle*, i.e. the form of supine: showing the supine stem.

The ways to obtain the stems from the principal forms are:

- *The present stem* can be found by dropping *-re* in the PRESENT INFINITIVE (Allen and Greenough, 1903, § 175).
- *The perfect stem* can be found from the third principal part: just remove *-ī*.
- *The supine stem* can be found by dropping *-um* in the supine i.e. the fourth principal form (Allen and Greenough, 1903, § 178).

Note that in Medieval Latin, often, instead of *iri* plus the first supine, *fore* plus the perfect participle is used to form the future passive infinitive. TODO: find a reference https://www.nationalarchives.gov.uk/latin/stage-2-latin/lessons/lesson-24-infinitives-accusative-and-infinitive-clause/

8.2.2 Formation of the present stem

8.2.3 Formation of the perfect stem

The PERFECT stem is not completely predictable from the PRESENT stem. Frequently observed alternations are listed below.

-*v*- suffix The etymological source of this suffix is not clear.

-*s*- suffix

Prolonged vowel

8.2.4 Conjugation classes

Depending on the way realization of the paradigm for a verb, Latin verbs are traditionally divided into four conjugations classes according to the thematic vowel of the stem: if the stem ends in *a*, it's a first conjugation verb (§ 8.3.3); if it ends in *e* then we have a second conjugation verb (§ 8.3.4); TODO. The conjugation classes however can't be reduced to the thematic vowel, since there are some contextual allomorphs that can't be explained synchronically in this way (§ 8.3.1, Table 8.3). There are a handful of irregular verbs (§ 8.7) that can't be inflected using rules pertaining to the four regular conjugations.

Another aspect of the inflectional behavior of the verb is whether it's deponent; this also has some implications on the argument structure (§ 8.6).

8.3 The finite paradigm

Due to morphophonological rules, morpheme division inevitably involves controversies. This section follows the analysis in Oniga (2014, chap. 14).

8.3.1 Marking of tense and mood

The contextual alternation of the tense and mood marker is listed in Table 8.3. They are subject to phonological rules and have allomorphs in different conjugation classes. A long vowel after the suffix, if any, is shortened before *-m*, *-r*, *-t*, and *-nt* due to vowel shortening in the final syllable before a consonant, and also before *-ntur* due to Osthoff's Law (§ 2.5.2; Oniga 2014, p. 130). The present indicative doesn't add a suffix

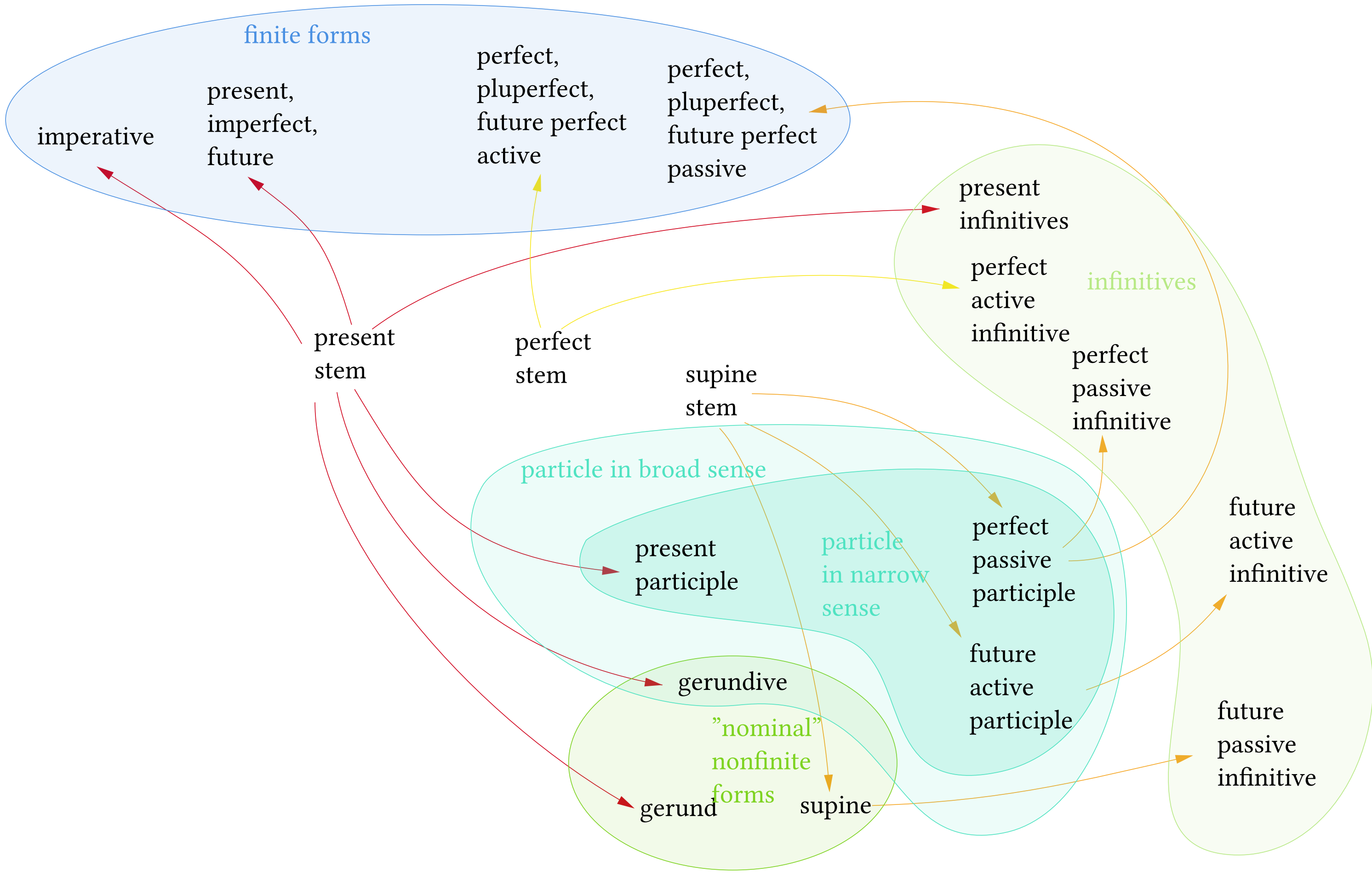

Figure 8.3: How to get all conjugation forms from the three stems

after the stem, so the thematic vowel at the end of the stem is directly exposed to the personal ending, and vowel changes in § 2.5 happen.

The perfect and pluperfect subjunctive suffixes are only used for the active voice. For the passive voice, periphrastic conjugation with the perfect passive participle is used.

Table 8.3: Tense and mood suffixes

Suffix	Mood and tense	Note
∅	Present indicative	With change on thematic vowel
-bā-, *-ebā-*	Imperfect indicative	Possibly shortened
-be-	Future indicative	For 1c, 2c; Allomorphs: *-b-*, *-bi-*, *-bu-*
∅	Perfect indicative	With its own personal endings
-erā-	Pluperfect indicative	Possibly shortened
-eri-	Future perfect indicative	Allomorph: *-er-* (1SG)
-ē-	Present subjunctive	Allomorphs: *-a-*, *-ā-*, *-e-*
-rē-	Imperfect subjunctive	Possibly shortened
-eri-	Perfect subjunctive	For active only
-issē-	Pluperfect subjunctive	For active only; possibly shortened

Box 8.2: Shortening or lengthening?

If we just restrict ourselves to the verbal paradigm, it may be attempting as well to consider *-ba-* as the indicative imperfect suffix, since *-bā-* does not outnumber it. But prolonging is rare in phonology: shortening, due to physiological motion control factors (some sound sequences are easier to pronounce), is more frequent. The same line of argumentation can be applied to justify the status of *-rē-* as the somehow canonical subjunctive imperfect suffix.

Perfect indicative active verb forms have their own set of personal endings (Table 8.4). The plural part of the personal endings still contains the usual *-mus*, *-tis*, *-nt* endings, and thus alternatively, we may analyze the verb endings for the PERFECT tense as the follows: the tense suffix is *-ī-* for first-person singular, third-person singular and first-person plural, *-is-* for second-person singular and second-person plural, and *-ēru-* for third-person plural; the first person singular ending is empty, and the second person singular ending is *-tī*, and the rest of personal endings are the same as those of other tenses.

Table 8.4: Personal endings for the PERFECT

Suffix	Person and number
-ī	1SG
-istī	2SG
-it	3SG
-imus	1PL
-istis	2PL
-ērunt	3PL

- The indicative:
  - Future:
    - For first and second conjugation verbs, the tense-mood morpheme is *-bi-*, except for first-person singular (which is *-b-*) and third-person plural (which is *-bu-*).
    - For third and fourth conjugation verbs, change stem-final stem.
  - Future perfect:
    - *-eri-* for all cases except first person singular.
    - *-er-* for first-person singular.
- The subjunctive:
  - Present: no suffixation, but there is regular change on the stem-final vowel:
    - For first conjugation verbs, .
    - For second conjugation verbs, $\rightarrow$ *eā-*.
    - For third conjugation verbs, $\rightarrow$ *ā-*.
    - For fourth conjugation verbs, $\rightarrow$ *iā-*.

    These alternations apply for both active and passive verbs, so they have nothing to do with polarity, and this is why I put them in this section.
  - Imperfect: *-rē-*, possibly shortened.

8.3.2 The personal ending

Possible personal endings are listed in Table 8.5. In the active voice, *-ō* is used with present indicative, future indicative (first and second conjugations only), future perfect indicative, and *-m* is used with imperfect indicative, future indicative (third and fourth conjugations only), pluperfect indicative, and the subjunctive mood regardless of tense. The PERFECT tense has its own personal endings: in the alternative analysis outlined in the last section, the first person perfect indicative ends with *-ī*, and the second person perfect indicative ends with *-tī*.

The ending *-re* is alternative form of second-person singular compatible with all non-periphrastic tenses and moods. If this personal ending is used, then the tense and mood marking is none. Note that the resulting verb form is the same as the infinitive principal part (§ 8.2.1).

The *-or* version of the passive first person ending is seen in present indicative and future indicative in the first and second conjugations; the latter has the deep form *-be-or*, then then is realized as *-bor* because of vowel deletion (§ 2.5.1).

Table 8.5: Personal endings

Suffix	Person and voice	Note
-ō, *-m*	active, 1SG	Depend on tense
-s	active, 2SG	
-t	active, 3SG	
-mus	active, 1PL	
-tis	active, 2PL	
-nt	active, 3PL	
-r, *-or*	passive, 1SG	*-or* for present and future indicative only
-ris	passive, 2SG	Allomorph: *-re* (without tense suffix)
-tur	passive, 3SG	Non-perfect tenses
-mur	passive, 1PL	Non-perfect tenses
-minī	passive, 2PL	Non-perfect tenses
-ntur	passive, 3PL	Non-perfect tenses

- The passive:
  - First-person singular:
    * *-r*: compatible with all tenses and moods, except the present indicative.
    * *-or*: present indicative. Also, note that the future indicative (first and second conjugations only) ending is *-bor*, which may be analyzed as .
  - Second-person singular:
    * *-ris*: compatible with all non-periphrastic tenses and moods.
    *

Box 8.3: Parsing a finite verb

Follow these steps to parse a finite verb:

- First see whether the aspect is perfect (by looking at the stem part) and the personal ending.
- Then TODO

8.3.3 The first conjugation

Attested thematic vowel alternations include:

- In first person singular present forms, the final *ā* is dropped, because the personal ending is *-ō* or *-or* and the thematic vowel is subject to vowel deletion.
- In active subjunctive *ā-* → *ē-*.

8.3.4 The second conjugation

Attested thematic vowel alternations include:

- In active indicative present forms, *ē* → *e*.

8.3.5 The third conjugation

The third conjugation has two subclasses: the thematic vowel-less case, and the *i*-thematic case.

8.3.6 Periphrastic conjugations

An auxiliary verb construction is a structure that contains one or more auxiliaries apart from the main verb, and yet is mono-clausal and is not a complement clause construction, and the auxiliaries are realizations of the verbal system surrounding the main verb, and not independent lexical verbs. Auxiliary verb constructions realizing grammatical categories that are usually realized by inflectional endings in a paradigm should be seen as a part of that paradigm, and therefore are known as periphrastic conjugations.

8.4 Uses of tenses and moods

8.4.1 The PRESENT

The

8.5 Non-finite forms

8.5.1 The infinitives

8.5.2 The gerund and participles

8.5.2.1 The gerund

The gerund is morphologically a neutral singular noun. The stem is formed by adding *-nd-* to the present stem; in other words, the accusative form of the gerund of the verb is obtained by removing the final *s* of the present active participle and adding *dum*. Note that the nominative case is missing – when a non-finite clause is required in the subject position, it's always an infinitive.

8.5.2.2 The present active participle

The present active participle (i.e. the present participle) is morphologically a third declension adjective (TODO: gender). The stem of the present active participle is obtained by adding *-nt* to the present stem. Equivalently, the nominative singular form – the citation form – is obtained by replace the *-re* ending of the present active infinitive by *-ns* (or in other words, add *-ns* to the present stem).

8.5.2.3 The perfect passive participle

The perfect passive participle (i.e. the perfect participle or the past participle) can be found by declining the neutral accusative past participle, i.e. the fourth principal part.

8.5.2.4 The future active participle

To get the future active participle (i.e. the future participle), add *-turus* to the supine stem.

8.5.2.5 The future passive participle

The future passive participle, or in other words the gerundive, can be obtained by declining the gerund as if it's an adjective (TODO: declension class); in other words the gerund is morphologically the singular neutral of the gerundive.

8.6 Deponent verbs

Deponent verbs are verbs that are in the passive voice realizationally, either with the corresponding morphological marking (1) or periphrastic marking (2), but still have active meaning. An observed tendency is that so-called "unaccusative" verbs seem to be deponent verbs in Latin, which means the passive voice essentially is the "non-agentive" voice (§ 9.1.2).

(1) [Confiteor] unum baptisma in remissionem peccatorum ...
confess-IND.PRES.PASS.1SG one-SG.ACC baptism-SG.ACC into/towards/about remission-SG.ACC sin-PL.GEN
'I confess one baptism for the remission of sins ...' (Nicene Creed)

(2) ... qui [locutus est] per prophetas ...
who speek.APRT-SG.NOM BE.IND.PRES.ACT.3SG through prophet-PL.ACC
'...who has spoken through the prophets ...' (Nicene Creed)

8.7 Irregular verbs

8.7.1 The verb *sum*

8.7.1.1 Overview

The verb *sum* has lots of uses in Latin grammar (§ 9.4.1), and its inflection is (unfortunately but expectedly) highly irregular. It's also defective: it has no passive forms, either finite or nonfinite. The principal parts (§ 8.2.1) are *sum*, *esse*, *fuī*, with the supine form being absent – usually replaced by the future active participle *futūrus*.

From the principal parts, we find the perfect stem is *fu-*, and the supine stem – if we insist on defining it – is the same, although the perfect passive participle is absent and so is the supine, and therefore the supine stem only appears in the future active participle.

The present stem is not well-defined: the second principal form *esse* doesn't have the regular infinitive ending *-re*, though we can roughly recognize something like *es-* or *e-*; the first principal form *sum* gives *su-* or *s-*. The two stems appear in the finite paradigm in an unpredictable manner, also with irregular though still recognizable endings. Besides *s-* and *es-*, there is also *fo-* seen in one variant of the future active infinitive (§ 8.7.1.2), which also appears in variants in the subjunctive active imperfect part of the finite paradigm.

8.7.1.2 The nonfinite paradigm

The only nominal form is the future active participle *futūrus*. The three active infinitives forms are all attested. The present active infinitive is *esse*. The perfect active infinitive is *fuisse*, regularly formed by the perfect stem *fu-*.

The future active infinitive can be regularly formed by adding *esse* to the future active participle, and therefore is *futūrum esse*. There is also a free variant *fore*.

8.7.1.3 The perfect system

The perfect forms – finite forms and the perfect active infinitive – of *sum* can be formed regularly (§ 8.3) according to the perfect stem *fu-*.

8.7.1.4 The imperative system

The present imperative system, which is known for reflecting the present stem, is formed regularly using *es-*: the singular second person present imperative is *es* and the plural second person present imperative is *este*.

8.7.1.5 The present system

The imperfect forms of *sum* are highly irregular, though patterns can still be found. In the indicative part (Table 8.6):

- The PRESENT forms show no pattern except the personal endings. Note that here *-m* instead of *-ō* is used for the first person singular form.
- The IMPERFECT forms are formed by adding the standard personal endings (*-m*, *-s*, *-t*, *-mus*, *-tis*, *-nt*) to *erā*, the vowel *ā* of which undergoes shortening according to rules in § 8.3.1.
- The FUTURE forms are formed by the same personal endings seen in the first and the second conjugations, although the tense marker isn't the same: the stem-tense marker complex is *er-* instead of the stem plus *-b-*.

Table 8.6: The indicative paradigm of *sum*

PRESENT	IMPERFECT	FUTURE
sum	*eram*	*erō*
es	*erās*	*eris*
est	*erat*	*erit*
sumus	*erāmus*	*erimus*
estis	*erātis*	*eritis*
sunt	*erant*	*erunt*

In the subjunctive paradigm (Table 8.7), we find that in the PRESENT system, the stem-tense marker complex is fused into *sī-*, and in the IMPERFECT system, the stem-tense marker complex is fused into *essē-* or *forē-*, both of which are then attached to the standard *-m*, *-s*, etc. personal endings, and the vowel shortening rule in § 8.3.1 works.

Table 8.7: The subjunctive paradigm of *sum*

PRESENT	IMPERFECT
sim	*essem, forem*
sīs	*essēs, forēs*
sit	*esset, foret*
sīmus	*essēmus, forēmus*
sītis	*essētis, forētis*
sint	*essent, forent*

8.7.2 The verb *faciō*

The verb *faciō* looks pretty regular regarding the endings, except for one thing: its *stem* alternates according to the voice.

8.8 Auxiliary verb constructions

An auxiliary verb, despite morphologically being a verb, is embedded into the TAM system of a language. The following aspects can be used to identify an auxiliary verb from a lexical one:

- *Standalone or not.* A lexical verb may appear on its own; an auxiliary verb never does this.
- *Scope and hierarchy compared with other TAM marking.* The relative order of complement-taking lexical verbs can be arbitrary; this is not true for auxiliary verbs.
- *Interplay with argument structure: voice.*
- *Interplay with argument structure: complement types.* A lexical verb usually imposes semantic confinements on the complement
- *Interplay with clause type.* Some auxiliary verbs never appear in certain types of non-finite constructions, because the TAM categories labeled by them are absent in these non-finite clauses.

Latin is usually perceived as a language with few analytic properties. Traditionally only *sum* and its forms are considered auxiliary verbs when they are used with the perfect passive participle to form perfect TAM constructions in passive voice. Some however argue that there are more auxiliary verbs.

Chapter 9

Verb frames

This chapter is mainly about verbs that don't take complement clauses as arguments. The phenomena discussed in this chapter mostly apply to complement clause constructions as well, but complement clause constructions have their own peculiarities (chap. 12).

9.1 Descriptive parameters

9.1.1 The argument-adjunct distinction

A clear complement-adjunct distinction – telling peripheral arguments from core arguments or oblique arguments – is hard to establish in Latin. Below I assess several parameters usually used to draw the distinction listed in Huddleston and Pullum (2002, § 4.1.2):

- *Content of clausal dependent.* Latin peripheral arguments do not necessarily have prepositions.
- *Easiness of topicalization, etc.* Latin is highly free-ordered and therefore all clause dependents can leave their base positions.
- *Obligatoriness.* Latin is also highly *pro*-drop, and even uncontroversial core arguments can be omitted.
- *Licensing by verb.* Oblique arguments are frequent in Latin, as is the case in English (consider *run away from* or *get into*).
- *Anaphora referring to core verb phrase.* Latin doesn't have systematic way to replace the core predicate (i.e. without adjuncts) by an anaphora.

Selection, licensing, and obligatoriness tests can still be used in more subtle ways (e.g. reference in discourse to identify omitted argument)

these criteria are however hard to use for a classical language.

Box 9.1: Argument and adjunct: directional construction

If a directional phrase obligatorily has the subject/object as its subject, then it has to be a complement and not an adjunct.

Thus, despite I'm fully aware that clausal dependents concerning place, instrument, mean, etc. may be licensed by both the argument structure of the verb and by clausal adjunct positions and may have clear structural differences in other languages (as in English), currently no distinction between the two cases is made. Following the traditional notion, the subject, several kinds of objects, the copular complement (Huddleston and Pullum (2002) calls it *predicative complement*) are identified as complements.

9.1.2 Argument as entity or situation

Verbs can be classified semantically into three subgroups, according to what their arguments refer to (Dixon 2005, Part B; Dixon 2010, § 18.5; and Dixon 2009, § 3.3):

1. Primary-A, which contains verbs that don't take arguments with meanings similar to those of complement clauses,
2. Primary-B, which are semantically **complement-taking**[1] and **lexical**, which have arguments that are semantically equivalent to complement clauses (but not necessarily syntactically coded as complement clauses) and have meanings more complicated then what's expected for grammatical items, and
3. Secondary, members of which have the same *meaning* of certain grammatical constructions in the verbal system, but not the same syntactic properties (for example, they may just take complement clauses instead of being auxiliary verbs).

The valency class of the verb is strongly related to but is not determined by the semantics of the verb.

9.2 Prototypical intransitive verbs

9.2.1 The MOTION type

9.3 Prototypical transitive verbs

With complement-taking verbs temporarily excluded, a prototypical transitive verb is more or less close to the AFFECT type, with an A argument which is the causer of the event

9.4 Copular verbs

9.4.1 The verb *sum*

It's also possible to use *sum* with an indirect object, and the meaning because 'something be to [someone]$_{\text{indirect object}}$'. In this case we get the possessive dative construction (Allen and Greenough, 1903, § 373).

TODO:

[1]Here the term means "semantically equivalent to a complement clause construction".

(1) Nono, utrum uti debeat metaphoricis vel
tenth whether at.any.rate owe-SUBJ.PRES-3SG.ACT metaphoric-PL.DAT/ABL or
symbolicis locutionibus.
symbolic-PL.DAT/ABL speech-PL.DAT/ABL
'Whether (it) at any rate uses metaphoric or symbolic speech.' (*Summa*, I q. 1 pr.)

9.5 Two place verbs

The case marking of the stimulus argument has some relations to its animacy (Pinkster, 2015, § 4.34).

9.6 Preverb

Roughly speaking, the preverbs are adverbs or preposition incorporated into the verb stem. The licensing condition of the preverb is highly restricted: it only takes place when a PATH category is licensed by the verb (which in Latin requires the event to be telic, and therefore the IMPERFECT tense and the like may be a problem for prefixation)

9.6.1 The prototypical dative construction

Latin also has two complement positions named as object: the indirect object and the secondary object. The indirect object is distinguished by the following grammatical properties:

- *Coding of semantic role*: in a AGT-type argument structure, the indirect object is usually the G argument. Intransitive clauses sometimes also have indirect objects, and an indirect object, in this case, is also a G argument.
- *Case marking*: indirect objects are always dative (§ 6.0.3).
- *Passivization*: indirect objects are always retained in passive clauses. They are never promoted to subjects in passivization.

9.7 Teaching and secondary object

The secondary object is distinguished by the following grammatical properties:

- *Coding of semantic role*: in an AGT-type argument structure that is always about information flowing, the T argument (i.e. the thing asked about or taught about) is the secondary object. The G argument (i.e. the person who is asked or taught) is the direct object. Sometimes the G argument is ablative, and in this case, there is only one accusative argument: the secondary object. Another place where secondary objects appear is clauses headed by a verb with a compounded accusative preposition.
- *Case marking*: secondary objects are always accusative.

- *Passivization*: secondary objects can be passivized, but much more rarely than direct objects.

-

The distributions of the secondary object and the indirect object are mutually exclusive. This means for ditransitive verbs of type GIVING, Latin shows a clear and strong tendency to identify the T argument with the monotransitive O, while for ditransitive verbs about teaching, the inverse is true.

Box 9.2: Comparison with English

It can be found that the Latin indirect object has more similarity with the English *to*-PP, which is also called the indirect object in some grammars, but not in Huddleston and Pullum (2002). The Latin indirect object differs from the English (accusative) indirect object in passivization. Since in Latin, verbs with AGT-type argument structure do not have alternation of complementation pattern – in English we have *give sth. to sb.* and *give sb. sth.*, while in Latin there is only the former one, but *to sb.* is replaced by a dative, (always with no preposition) – the G argument is identified with the E argument, and the T argument is identified with the P argument. In other words, in Latin, there is only the *John gave [goods]*$_T$ *to [charity]*$_G$ pattern: the double-object *John gave charity goods* pattern is absent.

Therefore, for typical ditransitive verbs, i.e. verbs like *give*, Latin shows a clear and strong tendency to identify the T argument with the monotransitive O, which is more typical than English[a], but for verbs with meaning of TEACH or ASK, there is also a clear and strong tendency to identify the G argument with the monotransitive O. The term *secondary object* is coined to cover this grammatical position.

[a] In English, in the *give sb. sth.* construction, it is the person i.e. the G argument that is passivized, while the T argument i.e. *sth.* cannot, though the latter is identified with monotransitive O according to other criteria.

9.8 (Change of) location

TODO: considering moving this section to the case section

Various semantic roles can be summarized as SOURCE, and the source clausal dependents – adjunct or complement – have the following properties. Note that we are dealing with a *group* of clausal dependents.

- *Coding of semantic role*:

- *Case marking*: a source argument is in the ablative case. It may come together with the prepositions *ex* or *ab*.

- *Passivization*: not available.

9.9 Psychological verbs

the psychological causer, i.e. the stimulus, is marked by the genitive, the target or benefactive or experiencer is marked by the dative.

Verbs about psychological activities often display unusual behaviors in world languages. The argument structure may be causative in which both the stimulus (which *causes* the experiencer to feel in a certain way) and the experiencer (who throws their emotion to a target) can be A-like. Further, in certain cases, the stimulus-causers originate from the stimulus-target, so that at least for some speakers, reverse binding as in *this rumor about herself frightened Mary* is acceptable (Hornstein and Motomura, 2002). The above two argument structures also have versions where the internal target argument or the internal experiencer argument is oblique: one unaccusative construction (*I'm anxious about …*) and one unergative construction (*I rejoice because of …*). We further have a possible impersonal construction where the experiencer and the stimulus form a small clause, and the subject is filled by a dummy pronoun. All the constructions have been attested in Latin (Giusti and Iovino, 2019).

Chapter 10

Constituent order

and a neutral order roughly described by Fig. 10.1 can be established. This order is due to both default information structure and argument structure, and order alternations on top of it due to information packaging are also well attested Prosody also plays a role in determining the constituent order.

Subject - direct object - oblique argument - adjuncts - source/goal - cognate object - verb

Figure 10.1: Latin neutral constituent order

The two competing factors – the peripheral nature of peripheral arguments, and direct object and certain oblique arguments being focused while others are not – lead to the neutral order in Fig. 10.1.

Box 10.1: Passivization in verbs about teaching

Can verbs taking accusative argument in (Pinkster, 2015, § 4.36) be passivized?

10.1 Configurationality in Latin

The word order in Latin poetry is more flexible than it is in prose. This is true in almost every language with a literary tradition. Indeed it is possible to compose English poetry in imitation of Latin (Allen and Greenough, 1903, § 600).

Available evidence supports the tradition in existing secondary literatures that Latin is thus better described as a discourse-configurational language, with multiple topicalization and focalization structures (Oniga 2014, p. 189; Danckaert 2017, p. 77; Devine and Stephens 2006, among others). Initial positions in clauses clearly bear information structure functions. Constituents that are able to move to the positions include almost everything: arguments, adverbials, the negator *non*, and also the verb (TODO: aux) (Allen and Greenough, 1903, § 598). TODO: is it possible for the main verb to move to the initial point only? Note that fronting of the verb may be used to focus the verb root or the *tense* (Allen and Greenough, 1903, p. 397); this means preposing of the verb is comparable to stressing the verb in English.

Apart from the information structure, prosody is also an important factor TODO

10.2 Broad focus clauses and the neutral order

Pragmatically unmarked sentences have the arguments as pragmatic topics and the rest of the VP are the focus: 'we already know [Baebius], [his army], and [Pinarius] (topics); the piece of new information is that [a transferring process happens involving the three] (focus)'. This kind of sentences, known as "broad scope focus" sentences in Devine and Stephens (2006, p. 15) because the scope of focalization is broad and not restricted to a single argument, demonstrates a constituent order which may be referred as the **neutral order** in Latin (Devine and Stephens, 2006, p. 79), although its frequency – without controlling the information structure – isn't significant higher than other constituent orders. This order clearly isn't a faithful representation of the argument structure, since the direct object – the argument that is supposed to be the closest one to the main verb – appears far from the main verb. The reason is likely to be that the "neutral order" also marks the aforementioned "unmarked" information structure, in which the arguments are by default topicalized.

Deviation from the "neutral" order may be divided into the following levels:

The usual typological classification of Latin as a SOV language, despite being misleading in suggesting a rigid base order, still makes sense in pointing out that indeed constituents that are "higher" are moved leftwards.

In periphrastic conjugation, the constituent order is subject + object + verb + *sum*. This may also show a mismatch between the dependency structure and the linear order, since if we consider the auxiliary *sum* to be the analytic counterpart of the inflectional suffix and has a higher position compared to the main verb (i.e. the participle), then since the subject and topicalized constituents usually appear on the left side, it also should appear on the left side compared with the participle under it. But the case may just be that the participle is focused and is moved leftward by default, just like the topicalized direct object, so the auxiliary then appears at the end of the clause in the neutral order.

10.3 Positioning of arguments

10.4 Positioning of the verb (without auxiliary)

Box 10.2: Position of the verb

The position of the verb involves a theoretical question: is its appearance away from the unmarked clause-final position due to phrasal movement (the verb root being moved to a new position, carrying all suffixal realizations of TP functional heads together with it), or is it due to being attracted by some sort of functional head (in this case the verb root is just like a head in head movement; similar mechanisms appear in, say, *on the top of the mountain* lies *a small village*)? Since this distinction is hard to test, I refrain from picking up one explanation.

10.5 Positioning of auxiliary and negation

One piece of evidence suggesting the grammatical status of *sum* is somehow different from a lexical verb is that its position has non-trivial interaction with the position of *non*. The negator *non* usually appears before the verb (Danckaert 2017, § 1.5, TODO: or aux?), and apparent violations seem to be constituent negation as opposed to sentential negation (Danckaert, 2017, p. 43).

10.6 On so-called postposing constructions

Whether postposing exists as an information structure marking device is still not completely clear. It's said that postposing is never used for emphasis (Allen and Greenough, 1903, p. 395), and apparent counterexamples are all "afterthoughts" TODO; but

The postponed subject is likely to be an afterthought, coindexed with a zero pronominal in the rest of the sentence before it Devine and Stephens (2006, p. 87), comparable to English *it kills three people, the wandering puma.*

10.7 Notes on some typologically rare constituent orders

10.8 Historical evolution

Without sentences in which the OV/VO alternation can be alternatively analyzed as topicalization, VO frequency no longer shows significant change as time went by, indicating a well-defined extended verb phrase (Danckaert, 2017, § 1.5, p. 29).

10.9 Information packaging constructions

10.10 Existential clause

In the existential construction, the *sum* verb always appears first (Allen and Greenough, 1903, p. 396).

10.11 Cleft construction

nequitia est quae te non sinit esse senem

10.12 Question formation

Chapter 11

Verb omission

Some constructions may be analyzed as "small clauses", which are smaller than usual clauses and lack real TAM marking; they are therefore not "real" clauses, but still appear frequently in Latin texts. (1) is a formula used frequently in Catholic Church and is an example of a small clause as a single and complete utterance. The fact that *gratias* is not in the nominative or the vocative case means (1) is neither a finite construction nor a single-noun phrase utterance.

(1) De-o grati-as
God-SG.DAT thank-PL.ACC
'(May) thanks be to God.'

Box 11.1: Why small clauses

Alternatively, small clause constructions can be analyzed as a full clause (for example, a copular clause) with the verb deleted. The main problem of this analysis is if this is true, we need to explain why prototypical transitive clauses never see their verbs omitted. It's therefore better to say that a copular clause is a small clause *plus* the copular, and that a giving or receiving ditransitive verb takes a small clause as its internal complement, which doesn't involve a questionable verb deleting process and also captures the intuition that a small clause is a full clause minus the verb.

Chapter 12

Complement clause constructions

12.1 Infinitives

Chapter 13

Relative constructions

13.1 General comments

Agreement properties The case of a relative pronoun is determined by its syntactic position in the relative clause, and *not* the case of the antecedent, though the number and gender categories are determined by agreement with the antecedent.

Chapter 14

Coordination and subordination

Latin coordination in the nominal system and the verbal system shows strong correspondence, with most conjunction words being shared by the two systems.

Chapter 15

Texts

Below are some examples of Latin texts, in an order from the easiest to the hardest, with remarks on their vocabulary and grammar.

15.1 *Aeneid*

15.1.1 Introduction

(1) Arma virumque cano, Troiae
weapon(N)-PL.ACC man(M)-SG.ACC=and sing-IND.PRES.1SG Troy-SC.GEN
qui primus ab oris Italiam,
REL.M.SG.NOM first-M.SG.NOM from shore(F)-PL.ABL Italy(F)-SG.ACC
fato profugus, Laviniaque venit
fate(N)-SG.ABL exiled-M.SG.NOM Lavinia-TODO=and go.to-IND.PRES.3SG
litora, multum ille et terris iactatus et alto vi superum saevae
shore(N)-PL.ACC
memorem Iunonis ob iram; multa quoque et bello passus, dum conderet

urbem, inferretque deos Latio, genus unde Latinum, Albanique patres, atque

altae moenia Romae.

'I sing weapons and a man, who was the first from the shores of Troy to Italy, was by fate exiled, and '

In (1), it should be noted that *arma* is in plural only. The *qui* clause is an example of the rule that the relative pronoun doesn't agree in case with the antecedent (§ 13.1). The copula is omitted in the *qui* clause.

15.2 Liturgy texts

15.2.1 Short formulae in the Roman Mass

Examples in this section are short formulae found in the Roman Mass in the order of their appearance. In (2, 3), *nomine* and *patris* are third declension nouns, while *spiritus* is a fourth declension noun.

(2) In Nomine Patris, et Filii, et Spiritus
in name-SG.ABL Father-SG.GEN and Son-SG.GEN and SPIRIT(M)-SG.GEN
Sancti.
holy-M.SG.GEN
'In the name of the Father, and of the Son, and of the Holy Spirit.'

(3) – Dominus vobiscum. – Et cum spiritu tuo.
Lord(M)-SG.NOM 2PL.ABL and with spirit(M)-SG.ABL your-M.SG.ABL
'– The Lord be with you. – And with your spirit.'

(4)

15.2.2 Nicene Creed

(5) Credo in unum Deum, Patrem
believe-IND.PRES.1SG in one-M.SG.ACC God(M)-SG.ACC father(M)-SG.ACC
omnipotentem,
omnipotent-M.SG.ACC
'I believe in one God, (the) omnipotent Father,'

(6) factorem caeli et terrae, visibilium omnium et
maker-
'maker of'

15.3 Vulgate bible

15.3.1 Excerpts in John 1

(7) in principio erat Verbum et Verbum erat apud Deum et Deus erat Verbum
in be.IMPF
'In the beginning' (John 1:1)

(8) omnia per ipsum facta sunt et
all-N.PL.NOM through DEM-ACC make.PPRT-N.PL.NOM be.IND.PRES.3PL and
sine ipso factum est nihil
without DEM.ABL make.PPRT-N.SG.NOM be.IND.PRES.3SG nothing.NOM
quod factum est
REL.N.3SG make.PPRT-N.SG.NOM be.IND.PRES.3SG
'All have been made through exactly this (i.e. the Word of Lord), and without exactly this, nothing that has been made has been made.' (John 1:3)

As an example, below I show how (8) can be parsed. First we can see a *et* dividing the sentence into two branches.

1. For the first branch, we know *omni-* is a quantifier meaning *all*, and morphologically it's a twin-termination third declension adjective; then from Table 5.2 and the fact that we are dealing with a third declension word, the ending *-a* means neutral and PL.NOM/ACC/VOC. The vocative case is of course impossible here.

2. The word *per* is a preposition taking an accusative object. *Ipsum* is a basic identity demonstrative, with the meaning of “exactly this”. Since it follows *per*, the ending *-um* here seems to be the accusative case marker, instead of a neutral nominative case marker.

3. The sequence *facta sunt* contains the indicative perfect 3pl copula *sunt*, and in *facta*, we see the supine stem *fact-* of the verb *faciō* ‘make’. The second fact means *facta* should be the perfect passive participle in a certain inflection form. Then *facta sunt*, collectively, is the indicative passive perfect 3pl form of *faciō*. (Here we are fortunate: it’s possible that *facta* and *sunt* get scattered to different places.) The *-a* ending can again be looked up for in Table 5.2: the possibilities are pl.nom/acc/voc – note that the first declension singular possibilities are excluded by the fact that *sunt* is in plural form. We expect *facta* to be nominative because it has to agree with the subject, which is always nominative and it turns to be possible.

4. Now we should link things together. The open ends are: the case of *omnia*, and the (3pl) subject of *facta sunt*. Then quite obviously, we find *omnia* should be in the subject position, and therefore everything works well.

5. We can also check gender agreements to make sure our reading is correct.

The second half is done in similar manners. The structure of the text looks like this:

> [[omnia]$_{\text{subject}}$ [per ipsum]$_{\text{instrument:PP}}$ [facta sunt]$_{\text{verbal complex}}$]$_{\text{coord}}$ et [[sine ipso]$_{\text{adverbial:PP}}$ [factum est]$_{\text{verbal complex}}$ [nihil [quod factum est]$_{\text{rel}}$]$_{\text{subject}}$]$_{\text{coord}}$

15.4 *Summa Theologiae*

15.4.1 Introduction

Thomas Aquinas, a theologian and philosopher, was recognized by the Catholic Church as one of the Doctors of the Church, usually known as *Doctor Angelicus* (doctor-sg.nom angle-adj-sg.nom, ‘Angelic Doctor’). *Summa Theologiae* (summary-sg.nom theology-sg.gen, ‘Summary of Theology’) or *Summa Theologica* (summary-sg.nom theology-adj-sg.nom.f), usually abbreviated as *Summa*, is probably his most known work. As its name implies, the book is a summary of Catholic theology, containing necessary information for beginning theological students and for arguing against frequently seen heresies. Apart from its great value within and out of Christianity, the book is a good example of what Medieval Latin looks like.

One salient feature of *Summa* is it’s written in a rather strict and even dull format. The work is divided into three Parts, each containing many Questions. Each Question (i.e. a topic, like “the nature and extent of this sacred doctrine”) is divided into several Articles (i.e. a specific question, like whether the sacred doctrine is a science). Each Article follows the following scheme:

1. First, a sentence like (9) indicating index of the Article within the Question it belongs to.

2. Then several Objections are raised to support an unorthodox idea, usually in subjunctive.

3. Then the accepted doctrine contrary to the above heretical claims is given (10).

4. Finally, a comment by Thomas on this Article (11).

(9) Ad primum sic procedit.
to first thus proceed-3SG.PASS
'(Our topic) goes to the first issue. (lit. It proceeds to the first.)'

(10) Sed contra est quod dicitur II
but contrarily be.IND.PRES.ACT.3SG what.R.SG.N say-IND.PRES-3SG.PASS two
ad Tim. III ...
to(TODO) NAME three
'But on the contrary there is what is said in 2 Timothy 3 ...' (*Summa*, I q. 1 s.c.)

(11) Respondeo dicendum quod ...
respond-IND.PRES-1SG.ACT
'I respond to what is said that ...'

Here we present a glossed Article in *Summa*: I q. 1 a. 9.

References

Edith Aldridge. Generative approaches to ergativity. *Language and Linguistics Compass*, 2(5):966–995, 2008.

Artemis Alexiadou and Florian Schäfer. *Instrument subjects are agents or causers.* Universitätsbibliothek Johann Christian Senckenberg, 2008.

Ti Alkire and Carol Rosen. *Romance languages: A historical introduction.* Cambridge University Press, 2010.

Joseph Henry Allen and James Bradstreet Greenough. *Allen and Greenough's New Latin grammar for schools and colleges: Founded on comparative grammar.* Ginn, 1903.

Irena Botwinik-Rotem and Arhonto Terzi. Greek and hebrew locative prepositional phrases: A unified case-driven account. *Lingua*, 118(3):399–424, 2008.

John Bowers. The syntax of predication. *Linguistic inquiry*, 24(4):591–656, 1993.

Guglielmo Cinque. *Adverbs and functional heads: A cross-linguistic perspective.* Oxford University Press, 1999.

Guglielmo Cinque. *The syntax of adjectives: A comparative study*, volume 57. MIT press, 2010.

James Clackson. *A companion to the Latin language.* John Wiley & Sons, 2011.

James Clackson and Geoffrey Horrocks. *The Blackwell history of the Latin language.* John Wiley & Sons, 2011.

Lieven Danckaert. Studying word order changes in latin. *Perspectives on historical syntax*, 169:233, 2015.

Lieven Danckaert. *The development of Latin clause structure: A study of the extended verb phrase*, volume 24. Oxford University Press, 2017.

Andrew M Devine and Laurence D Stephens. *Latin word order: Structured meaning and information.* Oxford University Press, 2006.

Anna Maria Di Sciullo. Decomposing compounds. *Skase Journal of theoretical linguistics*, 2(3):14–33, 2005.

RM Dixon. Basic linguistic theory. volume 3: Further grammatical topics, 2012.

Robert MW Dixon. *A semantic approach to English grammar.* OUP Oxford, 2005.

Robert MW Dixon. *Basic linguistic theory volume 1: Methodology*, volume 1. OUP Oxford, 2009.

Robert MW Dixon. *Basic linguistic theory volume 2: Grammatical topics*, volume 2. Oxford University Press on Demand, 2010.

Andrew R Dyck. The oxford latin syntax by harm pinkster. *Classical World*, 110(4): 575–576, 2017.

David Embick and Morris Halle. Latin inflections. *Paper delivered at Going Romance*, 2003.

David Embick and Morris Halle. On the status of stems in morphological theory. *AMSTERDAM STUDIES IN THE THEORY AND HISTORY OF LINGUISTIC SCIENCE SERIES 4*, 270:37, 2005.

Patrick Farrell. English verb-preposition constructions: Constituency and order. *Language*, pages 96–137, 2005.

Dianne Friesen. *A grammar of Moloko*. Language Science Press, 2017.

Jacopo Garzonio, Silvia Rossi, et al. Functional and lexical prepositions across germanic and romance. In *Continuity and Variation in Germanic and Romance*, pages 450–471. Oxford University Press, 2021.

Giuliana Giusti and Rossella Iovino. Psychological verbs as a vulnerable syntactic domain: A comparative study of latin and italian. *Lingua*, 223:29–45, 2019.

Paolo Greco. Latin accusativus cum participio: syntactic description, evidential values, anddiachronic development. *Journal of Latin Linguistics*, 12(2):173–198, 2013.

Nadine Grimm. *A grammar of Gyeli*. Language Science Press, 2021.

József Herman. *Vulgar latin*. Penn State Press, 2010.

Norbert Hornstein and Mitsue Motomura. Psych verbs, theta roles and reconstruction. In *Proceedings of 2002 LSK International Summer Conference*, volume 2, pages 39–58. Linguistics Society of Korea, 2002.

C. T. Huang. Hànyǔ dòngcí de tíyuán jiégòu yǔ qí jùfǎ biǎoxiàn (thematic structure of chinese verbs and their syntactic manifestations). *Yǔyán Kēxué (Language Sciences)*, 6(4):3–21, 2007.

Rodney Huddleston and Geoffrey K. Pullum. *The Cambridge Grammar of the English Language*. Cambridge University Press, 2002. doi: 10.1017/9781316423530.

Guillaume Jacques. Subjects, objects and relativization in japhug. *Journal of Chinese Linguistics*, 44(1):1–28, 2016.

Guillaume Jacques. *A grammar of Japhug*, volume 1. Language Science Press, 2021.

Julie Anne Legate. *Warlpiri: theoretical implications*. PhD thesis, Massachusetts Institute of Technology, 2002.

Jürgen Leonhardt. *Latin: Story of a world language.* Harvard University Press, 2013.

Yen-hui Audrey Li and Yuzhi Shi. Np as argument. *Y.-h. A. Li & A. Simpson (éds.), Functional structure (s), form and interpretation: perspectives from E ast A sian languages*, pages 3–27, 2003.

María Mare. Issues on word formation. the case of latin circum. *The Linguistic Review*, 35(1):121–159, 2018. This paper assumes that the main predicate in a clause whose main verb has *circum-* preverb is the preverb instead of the verb root, which is assumed to be adjoined to the light verb; this is not necessary for the conclusions of the paper; also, saying that the verb root is first adjoined to the light verb and then the complex is merged with the locative construction is not very different from saying that a "locationality" functional head selects the verb root as its complement and the locative construction as its specifier.

Jaume Mateu and Renato Oniga. Latin syntax in fifty years of generative grammar. *Catalan Journal of Linguistics*, 16:5–17, 2017.

Andrew McInnerney. *The argument adjunct distinction and the structure of prepositional phrases.* PhD thesis, University of Michigan, 2022. The paper claims that in English the post-verbal constituents have a flat structure. I'm not too sure about that, considering all the work done in Cartography.

Paul A Morris. Evidence of a configurational structure in meskwaki. *Proceedings of the Linguistic Society of America*, 3(1):64–1, 2018.

Stefan Müller. *Complex predicates: Verbal complexes, resultative constructions, and particle verbs in German.* CSLI publications, 2002.

Patrick M Niedzielski. *Clausal Syntax in the Sumerian Language.* PhD thesis, BA Thesis. Ithaca, NY: Cornell University, 2017.

Isabel Oltra-Massuet and Karlos Arregi. Stress-by-structure in spanish. *Linguistic Inquiry*, 36(1):43–84, 2005.

Maria Isabel Oltra Massuet. *On the notion of theme vowel: A new approach to Catalan verbal morphology.* PhD thesis, Massachusetts Institute of Technology, 1999.

Renato Oniga. *Latin: a linguistic introduction.* OUP Oxford, 2014.

Will Oxford. Inverse marking as impoverishment. In *Proceedings of WCCFL*, volume 34, pages 413–422, 2017.

Montserrat Pascual Pou. A syntactic analysis of instrumental prepositional phrases. *Catalan working papers in linguistics*, 9:053–68, 2001.

Waltraud Paul. *New perspectives on Chinese syntax.* De Gruyter Mouton, 2014.

Harm Pinkster. *The Oxford Latin Syntax: Volume 1: The Simple Clause.* Oxford University Press, 08 2015. ISBN 9780199283613. doi: 10.1093/acprof:oso/9780199283613.003.0004.

Harm Pinkster. *The Oxford Latin syntax: Volume II: The complex sentence and discourse.* Oxford University Press, 2021.

Jeffrey Punske. Three forms of english verb particle constructions. *Lingua*, 135:155–170, 2013.

Norvin Richards. Multiple case assignment and the english pseudo-passive. *A Schrift to Fest Kyle Johnson. Linguistics Open Access Publications*, pages 313–319, 2017.

Ana Paula Scher and Vitor Augusto Nobrega. Unifying neoclassical and stem-based compounds: a non-lexicalis approach. *Revista Linguíſtica*, 10(1):74–98, 2014.

Daniel Siddiqi. *Syntax within the word: Economy, allomorphy, and argument selection in Distributed Morphology*, volume 138. John Benjamins Publishing, 2009.

Daniel Siddiqi, Michael Barrie, Carrie Gillon, Jason D Haugen, and Eric Mathieu. *The Routledge Handbook of North American Languages.* Routledge, 2020.

Peter L Smith. *Greek and Latin roots: part I-Latin.* University of Victoria, 2016.

Peter Svenonius. Spatial P in English. In *Mapping Spatial PPs: The Cartography of Syntactic Structures, Volume 6*. Oxford University Press, 08 2010. ISBN 9780195393675. doi: 10.1093/acprof:oso/9780195393675.003.0004.

Michael L Weiss. *Outline of the historical and comparative grammar of Latin.* Beech Stave Press, 2009.

Michael Wilson, Maggie Baird, and Jonathan Pesetsky. The reversible core of object experiencer, location, and govern-type verbs. In *Proceedings of the 49th annual meeting of the North East Linguistic Society*, volume 3, pages 285–294. GLSA Amherst, 2020.

Susi Wurmbrand. The structure (s) of particle verbs, 2000.

Niina Ning Zhang. The syntax of english comitative constructions. 2007.